

A high-MAPK, low-WNT cell state drives metastatic dissemination in colorectal cancer

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Melanie Heinlein¹, Ginny X. Li², Noelyn Kljavin¹, Amanda R. Moore¹, Brian Biehs¹, Liming Tao³, Soufiane Boumahdi¹, Farnaz Mohammadi⁴, Minyi Shi⁵, Yuxin Liang⁵, Hartmut Koeppen⁶, Nicolò Riggi³, Robert Piskol⁴✉ & Frederic J. de Sauvage¹✉

Colorectal cancer (CRC), a leading cause of cancer-related mortality due to distant metastases, is largely driven by activating mutations in the WNT and mitogen-activated protein kinase (MAPK) pathways. Understanding the mechanism underlying the metastatic process is essential for developing effective treatments. Using serial in vivo orthotopic passaging, we developed an immunocompetent mouse model of metastatic CRC. Highly metastatic tumor cells exhibited chromosomal amplifications in MAPK pathway genes, resulting in increased MAPK pathway activity and suppression of WNT-associated transcriptional programs, including stem cell genes. Pharmacological inhibition of mutant KRAS^{G12D} led to a reduction in the MAPK-high–WNT-low transcriptional state and decreased both lung and liver metastases. Analysis of CRC patient data revealed that the metastatic gene signature associated with the MAPK-high–WNT-low state correlated with poorer survival outcomes. These findings underscore the plasticity of metastasis-initiating cells in CRC driven by the opposing roles of MAPK and WNT signaling, despite their synergy observed during colon tumorigenesis.

Colorectal cancer (CRC) is the third most common cancer worldwide and second in cancer-related deaths overall. Metastasis to distant organs, particularly the liver, is the primary cause of death in advanced CRC with a 5-year survival rate of <15% (ref. 1). Despite extensive characterization of the genetic events underlying CRC initiation, the cellular and molecular mechanisms driving metastatic dissemination and outgrowth are still poorly understood, representing a major barrier to therapeutic progress.

Lineage-tracing and functional studies have shown that Lgr5⁺ intestinal stem cells (ISCs) represent a cell of origin for CRC^{2–4}, whereby mutations in WNT (for example, *APC* and *CTNNB1*) and MAPK pathways (for example, *KRAS*-activating or *BRAF*-activating mutations) are critical for primary tumor initiation, whereas additional mutations in

tumor-suppressor pathways (for example, *TP53*) and transforming growth factor (TGF)β signaling (for example, *SMAD4* and *TGFBR2*) frequently contribute to metastatic progression by enhancing tumor adaptability and invasion^{5,6}. However, the role of ISCs in tumor maintenance and metastasis is more complex because Lgr5⁺ cells are dispensable for primary tumor growth. Depletion of these cells does not result in tumor regression, because a highly proliferative Lgr5[−] cell population drives tumor expansion and continuously replenishes the Lgr5⁺ compartment⁷.

Interestingly, recent studies have found that most disseminated CRC cells present in the circulation do not express Lgr5 (ref. 8) but markers such as LICAM⁹ or EMP1 (ref. 10). These disseminated metastatic cells demonstrate high plasticity, similar to the primary tumor, with Lgr5[−] metastasis-seeding cells regaining Lgr5 expression during metastatic

¹Department of Research Oncology, Genentech, South San Francisco, CA, USA. ²Roche Informatics, Hoffman-La Roche Canada, Mississauga, Ontario, Canada. ³Department of Cell and Tissue Genomics, Genentech, South San Francisco, CA, USA. ⁴Computational Sciences Center of Excellence, Genentech, South San Francisco, CA, USA. ⁵Department of Proteomic and Genomic Technologies, Genentech, South San Francisco, CA, USA.

⁶Department of Research Pathology, Genentech, South San Francisco, CA, USA. ✉e-mail: piskol.robert@gene.com; desauvage.fred@gene.com

outgrowth. This transition is critical for the survival and growth of metastases^{8,10} underscoring the complexity of CRC progression, where both *Lgr5*⁺ and *Lgr5*⁻ cells play essential, yet distinct, roles in tumor and metastases initiation and growth. These findings suggest that, although activating mutations in the WNT pathway are crucial for CRC tumor initiation, additional mechanisms may modulate WNT pathway activity as metastasizing cells transition between *Lgr5*⁺ and *Lgr5*⁻ states. Furthermore, they highlight the importance of cellular plasticity in metastatic progression, characterized by dynamic transitions between stem-like and non-stem-like states. Although activating mutations in the WNT and MAPK pathways are central to CRC pathogenesis^{6,11}, it remains unclear how these pathways interact to regulate cell-state transitions and metastatic competence. Current models often fail to recapitulate the full metastatic cascade in immunocompetent settings, limiting our understanding of tumor evolution in vivo.

Results

Generation of an immunocompetent mouse model of mCRC

To identify key drivers of CRC metastasis, we adapted an organoid-based model of metastatic CRC (mCRC)⁷ to create a system that enables mechanistic interrogation of metastatic progression in an immunocompetent setting. Using CRISPR–Cas9, we sequentially introduced mutations in *Apc*, *Kras*, *Trp53* and *Smad4* (AKPS) into small intestinal organoids derived from wild-type C57BL/6N mice (Extended Data Fig. 1a,b). Although these organoids reliably formed primary colon tumors and liver metastases in NSG mice, they failed to consistently metastasize to the liver in C57BL/6N mice, despite forming primary tumors (Fig. 1a,b and Extended Data Fig. 1c). To enhance metastatic competence, we employed in vivo serial orthotopic passaging by isolating liver metastases epithelial cells at a humane endpoint, expanding them in vitro and reinjecting them into successive cohorts of C57BL/6N mice (Fig. 1c). Over the course of five consecutive in vivo passaging rounds (Extended Data Fig. 1d), the organoids markedly enhanced their metastatic potential, with increased metastatic frequency and total number of metastases per liver (Fig. 1d,e) and lung (Extended Data Fig. 1e). The passage 5 (P5) organoid line m484 reliably showed the highest metastatic potential (Extended Data Fig. 1f–j), establishing a reproducible immunocompetent mCRC model, enabling deeper exploration of metastatic drivers and cell states in CRC.

Tumor-intrinsic changes increase the metastatic potential of in vivo passaged organoids

To identify tumor-intrinsic changes underlying the increased metastatic potential of in vivo passaged organoids, we first assessed whether additional somatic mutations were acquired in the highly metastatic P5 organoids. Whole-exome sequencing (WES) revealed 12–30 additional somatic point mutations across different passages compared to the parental (P0) organoids with little evidence for known oncogenic drivers of human CRC (Extended Data Fig. 2a,b), aligning with reports suggesting that metastasis may not necessarily depend on new driver mutations^{12–14} but rather arise through selection and acquisition of new traits¹⁵. In contrast, chromosomal instability analysis revealed a stepwise increase in chromosomal copy number alterations across passages with increased aneuploidy in the highly metastatic P5 compared to P1 organoids (Fig. 2a,b and Extended Data Fig. 2c), a trait often associated with advanced cancer stages¹⁶.

To determine the functional impact of copy number variation (CNV), we FACS sorted EpCAM⁺ tumor epithelial cells from tumors grown in vivo (Extended Data Fig. 2d) and performed RNA sequencing (RNA-seq) (Supplementary Table 1). CNVs correlated positively with transcriptional changes (Fig. 2c), with pathway analysis showing enrichment of DNA replication and cell-cycle programs in the metastatic P5 relative to non-metastatic P1 tumors (Fig. 2d, Extended Data Fig. 2e and Supplementary Table 2). Consistent with this, P5 organoids proliferated faster in vitro (Extended Data Fig. 2f) and formed more rapidly

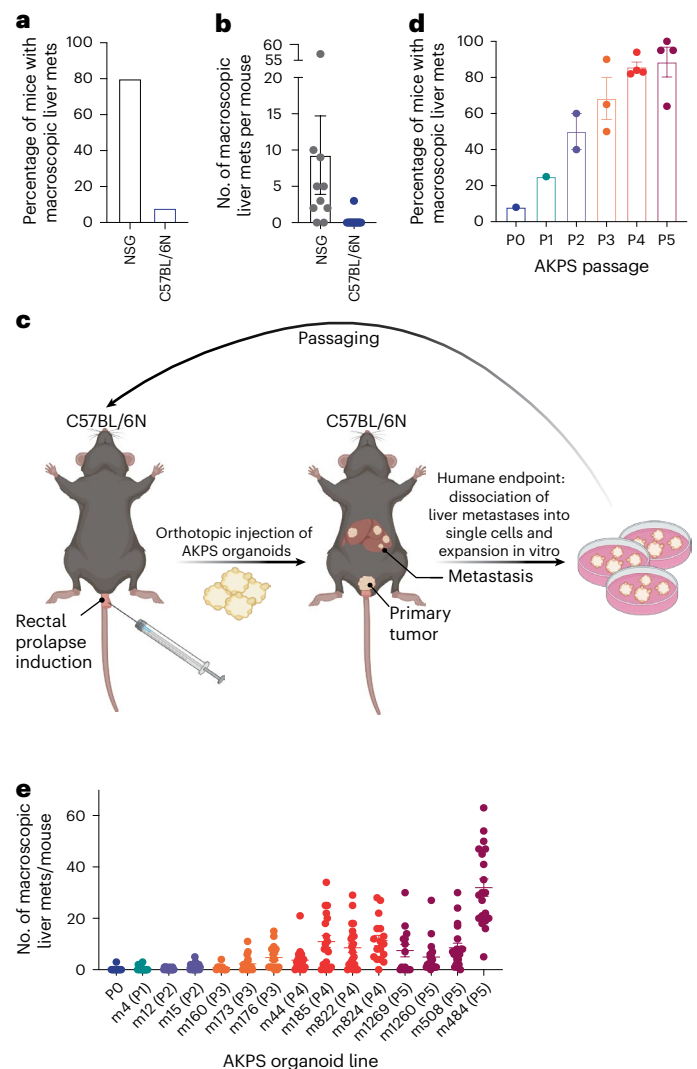


Fig. 1 | Generation of an immunocompetent mouse model of mCRC.

a, b, Percentage of mice with macroscopic liver metastases (**a**) and total number of macroscopic liver metastases per mouse (**b**) at humane endpoint in NSG ($n = 10$) and C57BL/6N ($n = 13$) mice. **c**, Schematic of the in vivo orthotopic passaging strategy used to generate an immunocompetent mouse model of mCRC. **d**, Percentage of C57BL/6N mice with macroscopic liver metastases across successive passage rounds at a humane endpoint. Each dot represents the percentage of mice with metastases for an individual organoid line. Note that data for P0 are the same as presented in **a, e**. **e**, Total number of macroscopic liver metastases per mouse at a humane endpoint. Each dot represents an individual mouse that was injected with the indicated organoid line: P0 $n = 13$; P1 (m4) $n = 16$; P2 (m12) $n = 18$; P2 (m15) $n = 20$; P3 (m160) $n = 10$; P3 (m173) $n = 17$; P3 (m176) $n = 17$; P4 (m44) $n = 18$; P4 (m185) $n = 19$; P4 (m822) $n = 22$; P4 (m824) $n = 17$; P5 (m1269) $n = 14$; P5 (m1260) $n = 19$; P5 (m508) $n = 20$; and P5 (m484) $n = 20$; n represents the number of individual mice. Note that data for P0 are the same as presented in **b**. In **b, d** and **e** data are presented as mean $\pm$ s.e.m. AKPS: *Apc*^{-/-}, *Kras*^{G12D}, *Trp53*^{-/-}, *Smad4*^{-/-}. Illustrations in **c** created in BioRender; Heinlein, M. <https://biorender.com/j44b227> (2025).

growing primary tumors in vivo, resulting in reduced overall survival in mice (Fig. 2e,f). Despite their metastatic burden, P5-injected mice were euthanized at a humane endpoint due to colon obstruction by the primary tumor. P1-injected mice occasionally survived nearly twice as long as P5-injected mice but rarely developed liver metastases (Extended Data Fig. 2g). Notably, tumors in both groups were of a similar size at the humane endpoint (Fig. 2g,h), suggesting that the acquired metastatic traits in P5 tumors were not driven by increased proliferation or tumor size but instead resulted from enhanced tumor cell

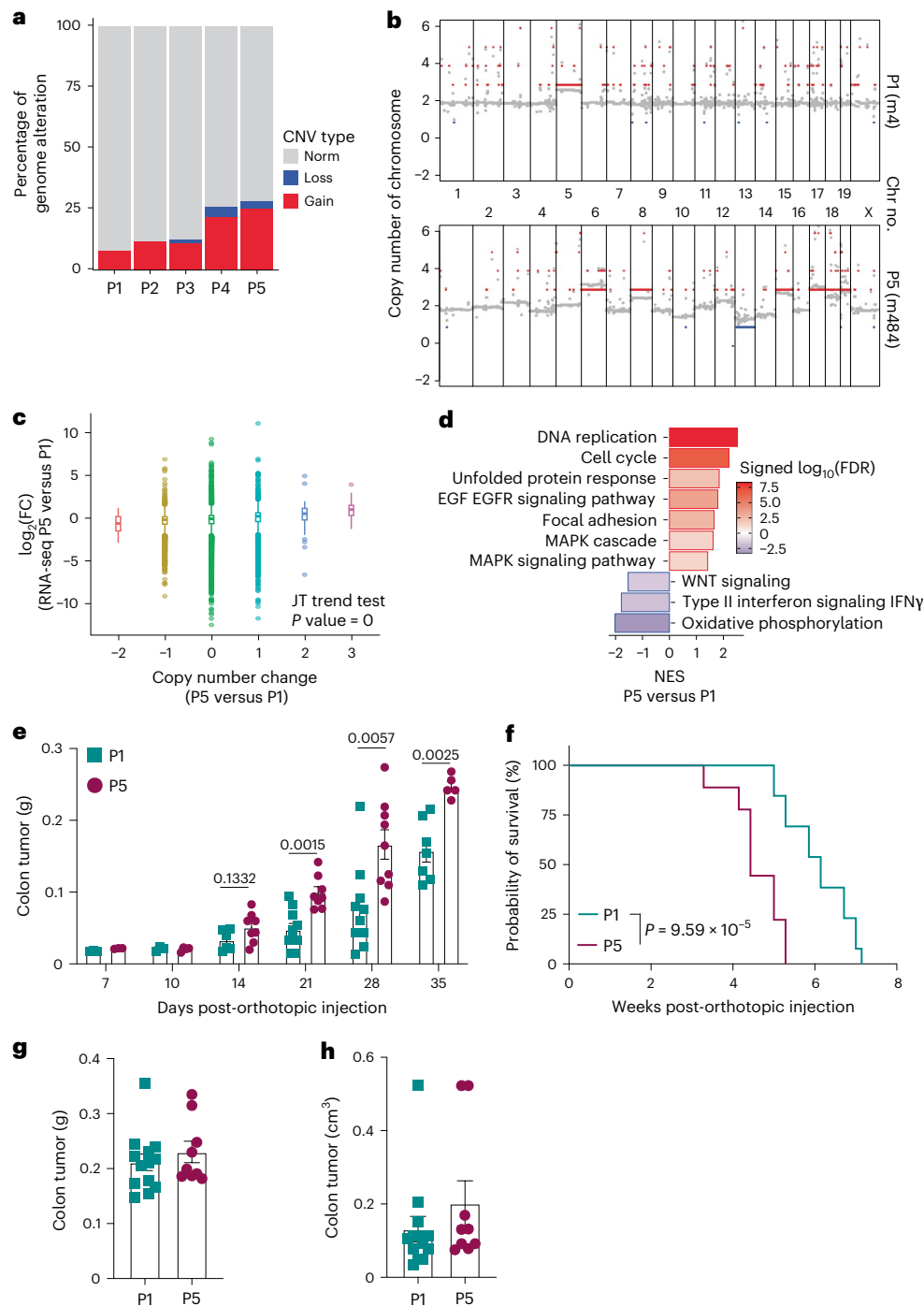


Fig. 2 | Changes in tumor-intrinsic properties enhance the metastatic potential of in vivo passaged organoids. **a**, Percentage of copy number alterations in the P1 (m4), P2 (m12), P3 (m176), P4 (m824) and P5 (m484) organoids compared to P0 (parental) organoids. **b**, Genome-wide copy number alterations in P1 (m4) (upper) and P5 (m484) (lower) across the whole mouse genome. Red shows copy number gain, blue copy number loss. **c**, The log₂(fold change) (log₂(FC)) of RNA-seq expression from sorted tumor epithelial cells of P5 (m484) (n = 3 mice) versus P1 (m4) (n = 4 mice) colon tumors plotted against chromosomal copy number changes between P5 (m484) (n = 1 replicate) and P1 (m4) (n = 1 replicate) organoids. Colored standard box plots depict distribution of expression change stratified by copy number change. Jitter dots represent genes. A one-sided Jonckheere–Terpstra (JT) test was used to assess trends in group medians from left to right. P = 0. Box plots show the center line representing the median (Q2), the box spans the interquartile range (IQR, Q1–Q3) and whiskers extend to Q1 – 1.5 $\times$ IQR and Q3 + 1.5 $\times$ IQR. **d**, Normalized gene set enrichment score of selected Wikipath gene signatures enriched in P5 (m484) tumor epithelial cells (red) or P1 (m4) tumor epithelial cells (blue). Color

gradients represent statistical significance. **e**, Primary colon tumor weight in grams at indicated timepoints post-orthotopic injection of the P1 (m4) and P5 (m484) organoids into the colon. Data are presented as mean $\pm$ s.e.m. Groups at each timepoint were compared using the Mann–Whitney U -test (unpaired, no Gaussian distribution, two tailed). P values are shown in the corresponding panels. For timepoints 7 d (P1 n = 3; P5 n = 3), 10 d (P1 n = 3 and P5 n = 4) and 21 d (P1 n = 10 and P5 n = 8), data from one independent experiment are shown; for timepoints 14 d (P1 n = 6 and P5 n = 8), 28 d (P1 n = 10 and P5 n = 9) and 35 d (P1 n = 7 and P5 n = 5), data are pooled from two independent experiments. **f**, Kaplan–Meier survival curve of mice orthotopically injected with P1 (m4) (n = 14) or P5 (m484) (n = 10) organoids. P = 9.59×10^{-5} was determined using log(rank test). **g**, **h**, Primary colon tumor weight in grams (**g**) and volume in cm³ (**h**) after resection at a humane endpoint: P1 (m4) n = 13; P5 (m484) n = 9. Data are presented as mean $\pm$ s.e.m. In **f**–**h** represents the number of individual mice used. In vivo experiments were repeated twice, except for the 7-d, 10-d and 21-d timepoints which were done once (**e**). chr, chromosome; NES, normalized enrichment score; norm, normal or unchanged.

dissemination from the primary site or improved colonization efficiency at secondary sites. Consistent with this hypothesis, hydrodynamic tail vein and intrasplenic injections showed no difference in the ability of non-metastatic P1 cells and metastatic P5 cells to generate liver or lung metastases (Extended Data Fig. 2h,i), confirming that the growth rate of metastases at the secondary site was similar between P1 and P5 cells.

Pathway analysis further revealed significant downregulation of type II interferon- γ (IFN γ) signaling in P5 tumor epithelial cells (Fig. 2d and Extended Data Fig. 2e), despite comparable levels of IFN γ in the tumor microenvironment (TME) (Extended Data Fig. 2j), indicating impaired pathway responsiveness. Consistent with this, P5 tumor cells exhibited reduced expression of IFN γ receptor subunits 1 and 2 (Extended Data Fig. 2k), accompanied by a marked absence of the chemokines CXCL9 and CXCL10 from the P5 TME (Extended Data Fig. 2l), which are critical for T and natural killer cell recruitment. In line with this, T cell infiltration was reduced in P5 primary tumors (Extended Data Fig. 2m). To assess additional immune evasive mechanisms, we examined major histocompatibility complex class I (MHC I) surface expression, which showed no significant differences between non-metastatic P1 and highly metastatic P5 organoids (Extended Data Fig. 2n). Last, to evaluate the overall contribution of immune surveillance to metastatic progression, P0, P2 and P5 organoids were orthotopically injected into immunodeficient NSG mice. Metastatic burden increased stepwise across passages (P0 < P2 < P5) (Extended Data Fig. 2o), suggesting that immune evasion contributes to the enhanced metastatic potential, but is not the sole driver implicating additional tumor-intrinsic mechanisms.

Differential enrichment of MAPK and WNT signaling pathways in metastatic tumors

Among the most differentially enriched pathways in the P5 tumor epithelium was the MAPK signaling cascade, despite both non-metastatic P1 and metastatic P5 organoids harboring the same activating *Kras*^{G12D} mutation (Fig. 2d and Extended Data Fig. 3a–c). Given the absence of additional activating point mutations in MAPK pathway genes (Extended Data Fig. 2b), we analyzed the WES data for aneuploidy in MAPK pathway genes¹⁷. This revealed an increase in copy numbers of genes encoding in *Kras*, *Braf*, *Raf1*, *Sos1*, *Mapk11*, *Mapk12*, *Mapk13* and *Mapk14*, corresponding to amplifications of chromosomes 6, 15 and 17 in the P5 organoids (Fig. 3a and Extended Data Fig. 3d). These amplifications were accompanied by increased expression of the affected genes (Fig. 3a) and higher expression of established MAPK target genes (*Spre1*, *Spre2*, *Spre4* and *Dusp4*) (Fig. 3b,c and Supplementary Tables 1 and 3), indicating elevated MAPK signaling activity. Consistent with an additional *Kras*^{G12D} copy (Fig. 3a), P5 organoids demonstrated increased sensitivity to the KRAS^{G12D}-specific inhibitor MRTX1133 (Extended Data Fig. 3e).

Conversely, the WNT pathway, which is required for maintaining Lgr5⁺ stem cells in the intestine, was notably downregulated in P5

tumor epithelial cells (Fig. 2d), despite the presence of an *Apc* truncation in both the P1 and the P5 organoids (Extended Data Fig. 3a,b and Supplementary Table 3). Canonical WNT target gene expression (*Lgr5*, *Smoc2* and *Nkd1*) was significantly decreased in metastatic P5 cells (Fig. 3d,e and Supplementary Tables 1 and 3) and comparison with stem cell signatures¹⁸ revealed a shift from an Lgr5⁺ crypt base columnar (CBC) state in P1 cells to an Lgr5⁺ regenerative stem cell (RSC) state in P5 cells (Extended Data Fig. 3f). Together, these data define a MAPK-high–WNT-low cell state associated with metastatic competence.

To test whether elevated MAPK signaling was sufficient to promote metastatic dissemination, we generated AKPS organoids with a *Kras*^{Q61R} mutation, which confers higher MAPK activity than codon 12 mutations due to enhanced GDP–GTP exchange, leading to a greater proportion of active, GTP-bound RAS^{19,20}. After orthotopic injection into the colon of C57BL/6N mice, approximately 60% of mice with AK^{Q61R}PS colon tumors developed macroscopic liver metastases, whereas none of the AK^{G12D}PS tumor-bearing mice did, despite comparable primary tumor sizes at the humane endpoint (Fig. 3f–h). Consistently, RNA-seq analysis revealed enrichment of epidermal growth factor (EGF) or MAPK signaling pathways and suppression of WNT signaling pathways in the more metastatic cells (P5 and AK^{Q61R}PS) compared to the non-metastatic cells (P1 and AK^{G12D}PS) (Extended Data Fig. 3g). Collectively, these results demonstrate that a MAPK-high–WNT-low state is a key driver of metastatic competence.

Pathway-specific chromatin accessibility dynamics reveal MAPK activation and WNT suppression in metastatic tumors

To further investigate the role of MAPK signaling in metastatic progression, we performed assay for transposase-accessible chromatin using sequencing (ATAC-seq) on tumor epithelial cells sorted from P1 and P5 colon tumors. Metastatic P5 tumor epithelial cells displayed extensive chromatin remodeling with increased accessibility at AP-1-associated transcription factor (TF) motifs, including FRA1, FOS and JUNB, key downstream effectors of MAPK signaling (Fig. 4a, Extended Data Fig. 4a–c and Supplementary Tables 4 and 5). In addition, motifs for TEAD, the transcriptional partner of YAP²¹, were enriched, consistent with expression of a YAP-driven intestinal gene signature²² in P5 tumor cells (Extended Data Fig. 4c–d and Supplementary Table 5). Conversely, motifs associated with WNT-related TFs, including TCFL2 and LEF1, showed markedly reduced accessibility in metastatic cells (Fig. 4a, Extended Data Fig. 4e–g and Supplementary Table 5). Treatment of P5 organoids with a panel of epigenetic inhibitors did not induce convergence of the P5 chromatin accessibility profile toward a P1-like state, suggesting that the pro-metastatic chromatin program was resistant to single-agent epigenetic modulation (Extended Data Fig. 4h).

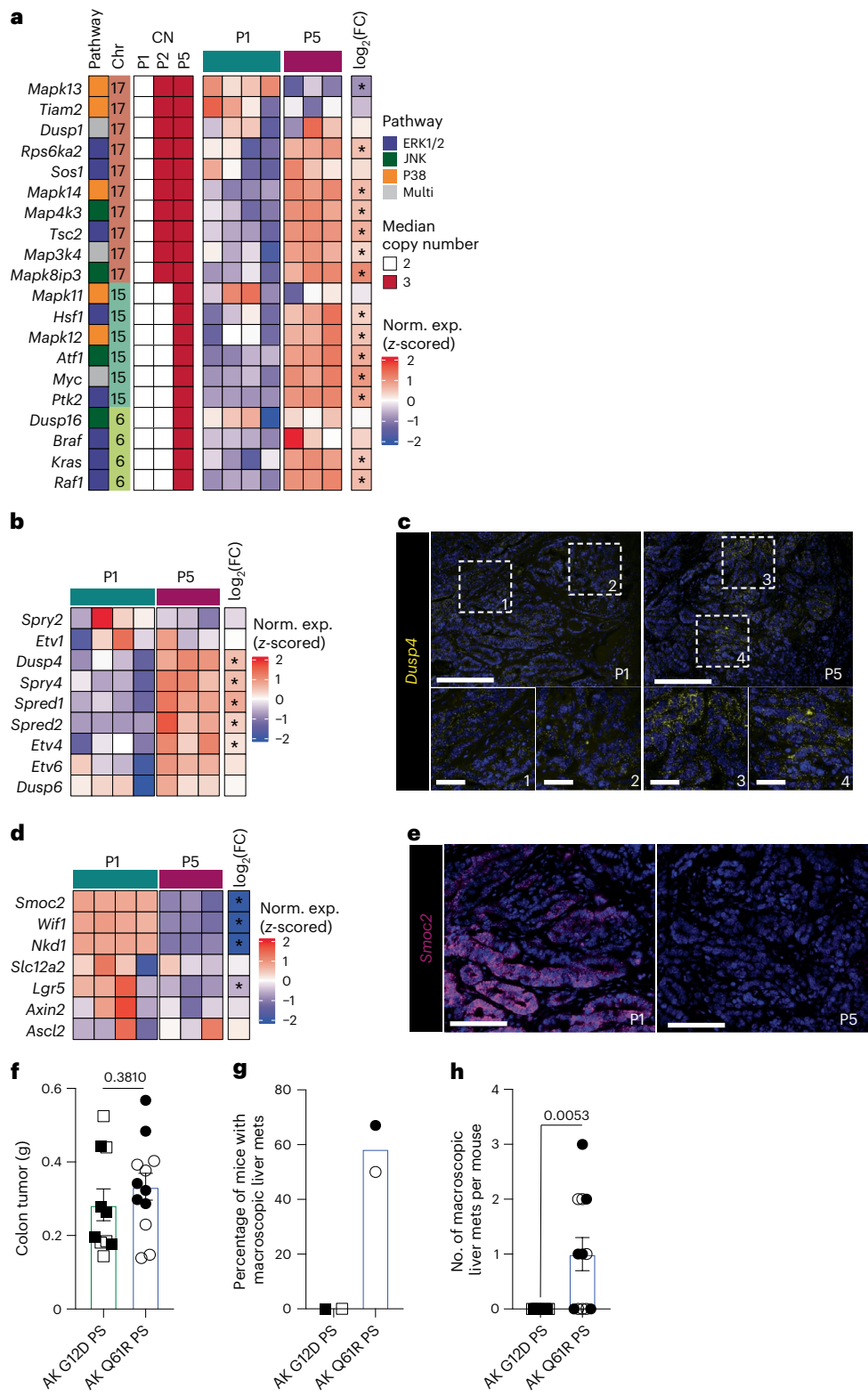
To link chromatin accessibility to transcriptional output, we integrated ATAC-seq and bulk RNA-seq data from matched P1 and P5 tumors using Binding and Expression Target Analysis (BETA)²³.

Fig. 3 | Differential enrichment of MAPK and WNT signaling pathways in metastatic tumors. **a**, Copy number (CN) changes of MAPK-related pathway genes¹⁷ on chromosomes 6, 15 and 17 in P1 (m4), P2 (m15) and P5 (m484) organoids relative to P0 (parental) organoids ($n = 1$ replicate per organoid line), together with normalized expression (norm. exp.; z-scored) and $\log_2$ (FC) of the same genes in tumor epithelial cells sorted from P5 (m484) and P1 (m4) colon tumors 5 weeks post-orthotopic injection. The asterisks indicate significant fold-changes (false discovery rate (FDR) ≤ 0.05). The exact FDR values for all fold-changes are provided in Supplementary Table 1: P1 (m4) $n = 4$; P5 (m484) $n = 3$ mice. **b**, Normalized expression (z-scored) and $\log_2$ (FC) of downstream MAPK pathway target genes in tumor epithelial cells sorted from P5 (m484) and P1 (m4) colon tumors 5 weeks post-orthotopic injection. The asterisks indicate significant fold-changes (FDR ≤ 0.05). The exact FDR values for all fold-changes are provided in Supplementary Table 1: P1 (m4) $n = 4$; P5 (m484) $n = 3$ mice. **c**, Representative images of *Dusp4* in situ hybridization (ISH) on P1 (m4) colon tumors (left) and P5 (m484) (right) colon tumors 5 weeks post-orthotopic injection. Scale bar: 100 μ m (main) and 25 μ m (insets). Sections are

from $n = 3$ mice per group. **d**, Normalized expression (z-scored) and $\log_2$ (FC) of downstream WNT target genes in tumor epithelial cells sorted from P5 (m484) and P1 (m4) colon tumors 5 weeks post-orthotopic injection. The asterisks indicate significant fold-changes (FDR ≤ 0.05). The exact FDR values for all fold-changes are provided in Supplementary Table 1: P1 (m4) $n = 4$; P5 (m484) $n = 3$. **e**, Representative images of *Smoc2* ISH on P1 (m4) colon tumors (left) and P5 (m484) colon tumors (right) 5 weeks post-orthotopic injection. Scale bar: 100 μ m. Sections are from $n = 3$ mice per group. **f–h**, Primary colon tumor weight in grams (**f**), percentage of mice presenting with macroscopic liver metastases (**g**) and total number of macroscopic liver metastases per mouse at the humane endpoint (**h**). Data were pooled from two independent experiments (open symbols, repeat 1; filled symbols, repeat 2; AK^{Q61R}PS $n = 6$ mice per repeat. AK^{G12D}PS $n = 5$ mice per repeat). Groups were compared with unpaired Student's *t*-test (two tailed). *P* values are shown in the corresponding panels. In **f** and **h**, each point represents a mouse. Data are presented as mean $\pm$ s.e.m. In **g**, the data points represent the percentage of mice within each repeat experiment. Data are presented as the mean. Norm. exp., Normalized expression.

Motif-specific BETA revealed that TCF–LEF motifs were associated with a WNT-driven transcriptional program in P1 cells, including *Wif1*, *Nkd1* and *Smoc2* (P1 WNT BETA signature), whereas AP-1 motifs drove a MAPK-associated transcriptional program in P5 tumor epithelial cells (P5 MAPK BETA signature) (Fig. 4b and Supplementary Table 3). Among P5-specific target genes was *Emp1* (Fig. 4b), a marker of high-risk CRC cells with invasive and metastatic potential^{10,24}. Consistently, metastatic P5 cells displayed significantly higher chromatin accessibility at the *Emp1* locus near MAPK-associated TF-binding sites,

suggesting that MAPK signaling drives *Emp1* expression (Fig. 4c). ENCODE chromatin immunoprecipitation sequencing (ChIP-seq) datasets confirmed binding of AP-1 family members, including BATF, JUNB and FOS, at the *EMP1* promoter region in multiple human cell lines (Extended Data Fig. 4i), strengthening a mechanistic link between MAPK pathway activation and *EMP1* upregulation. Conversely, the *Smoc2* locus showed significantly reduced chromatin accessibility at predicted TCF–LEF-binding sites in P5 cells (Fig. 4d), indicating a shift in cell identity from a WNT-high stem state (P1) to a MAPK-high state



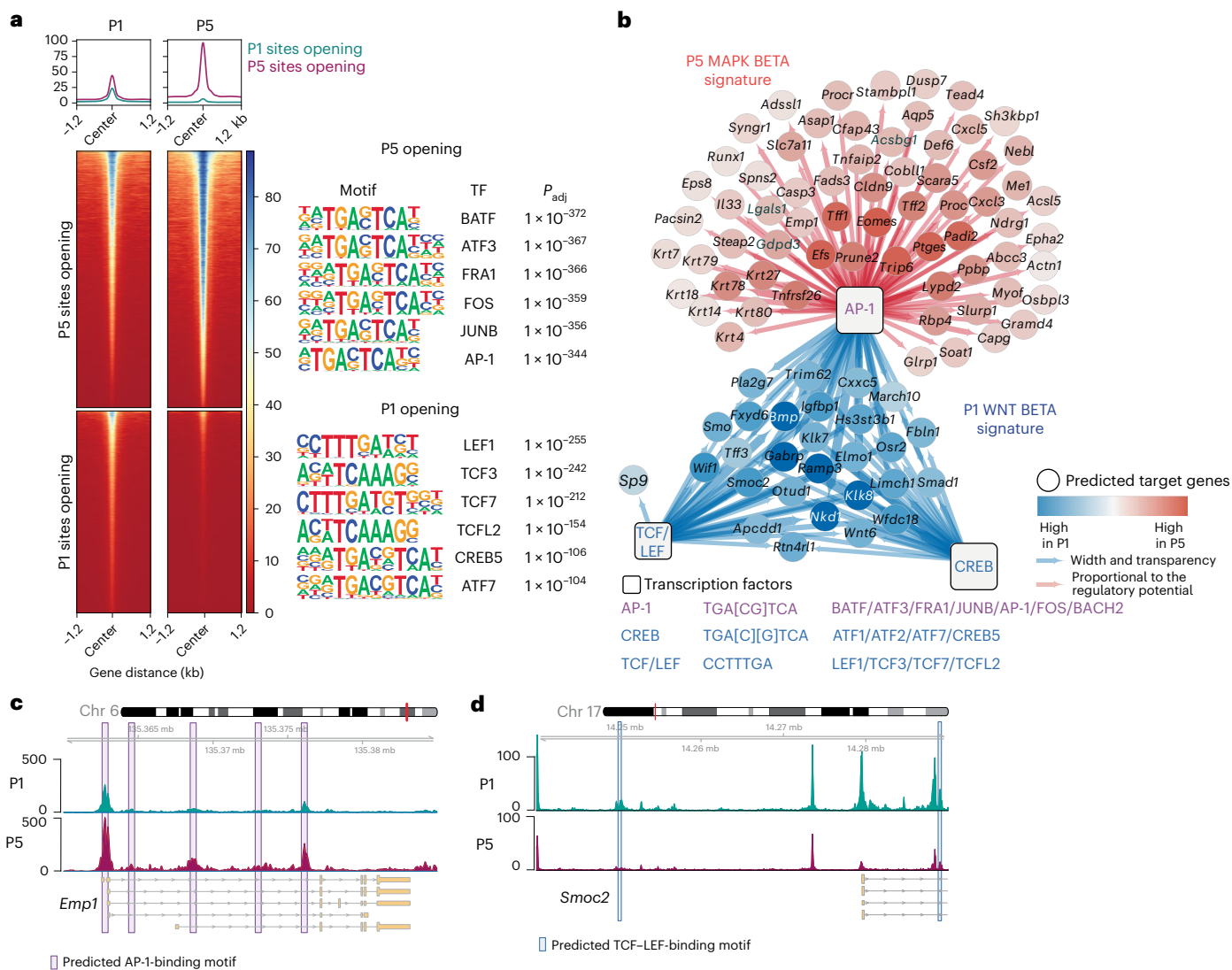


Fig. 4 | Pathway-specific chromatin accessibility dynamics reveal MAPK activation and WNT suppression in metastatic tumors. a, ATAC-seq on sorted tumor epithelial cells from P5 (m484) and P1 (m4) colon tumors 5 weeks post-orthotopic injection. Histograms and tornado plots (left) show normalized library size and averaged group counts around the transcriptional start site (center; 1,000 bp upstream, 400 bp downstream) for regions with increased accessibility in P5 (maroon) or P1 (teal). Logo plots (right) show the top six enriched TF motifs in P5 (top) and P1 (bottom) opening regions. Adjusted P values (P_{adj}) are derived from one-sided hypergeometric tests with Benjamini–Hochberg FDR. **b**, Regulatory transcriptional network showing BETA-predicted gene targets of the most enriched TF motifs in P5 (m484) and P1 (m4) colon tumor epithelial cells. AP-1-associated TFs (motifs: TGA[CG]TCA) were predicted to influence the transcription of genes in P5 (red; P5 MAPK BETA signature) and to a lesser extent in P1 (blue). WNT-associated TFs (motifs: CCTTTGA) were predicted to only influence the transcription of genes in P1 in conjunction with AP-1 and CREB TF motifs (motifs: TGA[C][G]TCA) (blue; P1 WNT BETA signature). Circles represent

target genes, with fill color indicating the magnitude of differential expression based on paired RNA-seq data on sorted tumor epithelial cells from P5 (m484) and P1 (m4) colon tumors. Arrow width and transparency are proportional to regulatory potential as determined by rank product analysis (RNA-seq: P1 (m4) $n = 4$; P5 (m484) $n = 3$). **c**, Coverage plots showing the library size normalized and group-averaged ATAC-seq counts at the *Emp1* locus in P1 (m4) (upper) and P5 (m484) (lower) tumor epithelial cells. The purple boxes indicate regions with increased accessibility in P5 corresponding to predicted MAPK-associated TF-binding sites (AP-1 motif: TGA[CG]TCA). The genome track above indicates the location of the gene on the chromosome (red line). **d**, Coverage plots showing the library size normalized and group-averaged ATAC-seq counts on the 5'-end of *Smoc2* in P1 (upper) and P5 (lower) tumor epithelial cells. The blue boxes indicate regions with increased accessibility in P1 corresponding to predicted WNT-associated TF-binding sites (TCF-LEF motif: CCTTTGA). The genome track above indicates the location of the gene on the chromosome (red line). In **a–d**, ATAC-seq: $n = 4$ mice per group.

(P5). We hypothesized that the shift in the transcriptional landscape was critical for the metastatic potential of P5 tumor epithelial cells.

KRAS^{G12D} inhibition reverts the metastatic transcriptional cell state to a non-metastatic cell state

Previous reports have shown that most CRC metastasis-seeding cells are Lgr5⁺ (refs. 8,10). To investigate whether MAPK signaling drives tumor cell dissemination, we treated mice with P5 allografts with the KRAS^{G12D}-specific inhibitor MRTX1133 (ref. 25). After confirming

effective inhibition of MAPK-specific downstream targets in AKPS organoids (Extended Data Fig. 5a,b), mice were treated early during metastatic progression with MRTX1133 for 14 d (Fig. 5a). Although primary tumor size was noticeably reduced (Fig. 5b), no macroscopic liver metastases and only few lung metastases were found (Fig. 5c,d), highlighting the critical role of high-MAPK pathway activity in driving metastasis.

Building on our findings that MAPK-associated TFs regulate *Emp1* expression (Fig. 4b,c), we examined the impact of MAPK inhibition on

this transcriptional state. Short-term MRTX1133 treatment (3 d) significantly reduced *Emp1* and *Dusp4* expression in tumor epithelial cells (Fig. 5e), confirming its dependence on MAPK activity and disruption of the metastatic transcriptional program.

Consistent with this, bulk RNA-seq of sorted EpCAM⁺ colon tumor epithelial cells from MRTX1133-treated mice (Extended Data Fig. 5c–f and Supplementary Table 1) revealed a shift of the transcriptional state of the P5 samples toward that of non-metastatic P1 samples (principal component (PC)1 dimension) (Fig. 5f). This shift was driven by differences in cell-cycle-related pathways (P5 and vehicle-treated samples) and WNT-associated pathways (P1 samples) (Extended Data Fig. 5g and Supplementary Table 6). Variations in the PC2 dimension were influenced by MAPK-related pathways (P1, P5 and vehicle) and metabolic pathways (MRTX1133 treated) (Extended Data Fig. 5h and Supplementary Table 6). MAPK target genes were significantly downregulated after 3 d of treatment (Fig. 5g). After 16 d, downregulation persisted, although to a lesser extent, likely reflecting the emergence of resistant cells (Fig. 5g). In contrast, known stem cell and WNT target genes were upregulated on KRAS^{G12D} inhibition (Fig. 5g), underscoring the inverse relationship between the level of MAPK and WNT activity described above.

Analysis of the P1-specific and P5-specific BETA signatures confirmed that KRAS^{G12D} inhibition reversed the metastatic MAPK-high–WNT-low state (Fig. 5h and Extended Data Fig. 5i–k) and reverted P5 organoids from an RSC-like to a CBC-like state (Extended Data Fig. 5l,m). Notably, KRAS^{G12D} inhibition also led to a significant decrease in the YAP-dependent transcriptional signature of the metastatic P5 cells (Extended Data Fig. 6a,b). To investigate whether the reactivation of WNT signaling was YAP dependent, we treated P5 organoids with MRTX1133, a pan-TEAD inhibitor GNE-7883 (ref. 26) (to block YAP/TEAD-mediated transcription) or a combination of both. Interestingly, pharmacological inhibition of TEAD signaling alone or in combination with KRAS inhibition did not further modulate MAPK or WNT target gene expression beyond KRAS inhibition alone (Extended Data Fig. 6c), indicating that WNT reactivation is primarily driven by MAPK suppression.

To investigate the impact of KRAS^{G12D} inhibition on late-stage mCRC, mice with established liver metastases were treated with MRTX1133 for 3 or 14 d (Fig. 5i). Although primary tumor size remained largely unchanged (Fig. 5j), EpCAM⁺ tumor epithelial cells were significantly reduced (Fig. 5k), accompanied by increased MHC1 expression (Extended Data Fig. 6d,e). Histological analysis showed thinning of the tumor epithelium with distinctly larger luminal

spaces, a notable expansion of fibroblasts (Fig. 5l) and influx of T cells (Extended Data Fig. 6f). Liver metastases showed minimal reduction in number, but similar flattening of the tumor epithelium and fibroblast expansion compared to the primary tumor (Fig. 5m,n). Notably, lung metastases were highly sensitive to treatment, showing fast and sustained reduction in metastatic burden and near-complete loss of tumor epithelial cells (Fig. 5o,p). In addition, MAPK target gene expression (*Dusp4* and *Emp1*) was reduced through day 14, whereas WNT target gene expression (*Smoc2*) increased at both metastatic sites 3 d after treatment (Extended Data Fig. 7a–d) further supporting the inverse relationship between MAPK and WNT activity described above, even in distant metastases. Analysis of an independent single-cell RNA-seq (scRNA-seq) dataset of tumor epithelial cells isolated from liver and lung metastases from NSG mice orthotopically injected with AKVPSL organoids⁷ revealed site-specific baseline differences in MAPK and WNT activity, with liver metastases showing higher *Smoc2* expression and stronger enrichment for the P1 WNT BETA signature compared to lung metastases (Extended Data Fig. 7e–i). This was corroborated by nuclear localization of β -catenin, a transcriptional effector of WNT signaling, in liver metastases (Extended Data Fig. 7j) in the same study. These findings suggest that differential baseline pathway activity contributes to site-specific responses to KRAS inhibition.

In summary, these data demonstrate that KRAS^{G12D} inhibition had limited impact on primary tumor growth but significantly reduced tumor cell dissemination, potentially by disrupting cellular plasticity and changing the transcriptional state of highly metastatic cells (MAPK-high–WNT-low) into a less-metastatic state (WNT-high–MAPK-low).

A MAPK-high–WNT-low transcriptional state is associated with shorter overall survival in patients with CRC

Last, we assessed the clinical relevance of these findings in human CRC. We hypothesized that tumors with high MAPK activity, as indicated by elevated expression of canonical MAPK downstream target genes, together with a loss of WNT activity, would correlate with shorter survival due to more aggressive disease progression. Analysis of RNA-seq data from treatment-naïve patients with colorectal tumors in the AVANT²⁷ and CALGB²⁸ cohorts revealed that patients with a high-MAPK–low-WNT score had significantly shorter overall survival, whereas a low-MAPK–high-WNT was associated with better survival in both trials (Fig. 6a,b, Extended Data Fig. 8a,b and Supplementary Table 7). Although KRAS mutant tumors displayed higher expression of MAPK target genes compared with KRAS wild-type tumors in the AVANT

Fig. 5 | KRAS^{G12D} inhibition reverts the metastatic transcriptional cell state to a non-metastatic cell state. **a**, Schematic for MRTX1133 treatment in mice orthotopically injected with P5 (m484) organoids into the colon. Mice were treated with vehicle or MRTX1133 (30 mg kg^{−1} intraperitoneally (i.p.) twice a day) for 14 d starting at 3 weeks post-orthotopic injection (inj.). **b–d**, Primary colon tumor weight (**b**), total macroscopic liver metastases (**c**) and total macroscopic lung metastases (**d**) after treatment ($n = 5$ mice per group). **e**, Representative images of *Dusp4* (yellow, left) and *Emp1* (magenta, middle) ISH and merged images (DAPI blue, right) of P5 colon tumors treated with vehicle (top) or MRTX1133 for 3 d (bottom), 5 weeks post-orthotopic injection. Scale bar: 100 μ m. Sections are from $n = 3$ mice per group. **f**, Principal component analysis plot showing the distribution of RNA-seq samples on sorted colon tumor epithelial cells from P1 (m4) (teal; $n = 4$) and P5 (m484) (maroon; $n = 3$) (from Fig. 3) and P5 (m484) treated with vehicle (3 d; black; $n = 4$) or MRTX1133 (3 d; yellow; $n = 4$ or 16 d; blue; $n = 4$). Each point represents a mouse. Axis labels show the percentage of variance explained. **g**, Normalized expression (z-scored) and log₂(FC) of downstream WNT and MAPK target genes in tumor epithelial cells from P5 (m484) tumors treated with vehicle (3 d) or MRTX1133 (3 d or 16 d). The asterisks represent fold-changes that are significant (FDR ≤ 0.05). The exact FDR values for all fold-changes are provided in Supplementary Table 1 ($n = 4$ mice per group and timepoint). **h**, Enrichment plot showing the association of the P5 MAPK

BETA signature (maroon) or P1 WNT BETA signature (teal) between colon tumors treated with MRTX1133 (16 d) or vehicle (3 d). The ranked list of genes is shown below in gray. Two-sided NES with Benjamini–Hochberg correction. **i**, Schematic of MRTX1133 treatment, with dosing initiated 5 weeks post-orthotopic injection of P5 (m484) into the colon for either 3 d or 14 d (30 mg kg^{−1}, i.p., twice a day). **j**, Primary colon tumor weight after resection at the indicated timepoints (vehicle 3 d $n = 5$; MRTX1133 3 d $n = 6$; control at 35 d $n = 4$; MRTX1133 14 d $n = 6$). **k**, EpCAM⁺ cells $\times 10^7$ per g of colon tumor tissue as determined by FACS analysis on P5 (m484) colon tumors at indicated timepoints and treatments ($n = 4$ mice per group and timepoint). **l**, Representative hematoxylin and eosin (H&E)-stained sections of P5 (m484) colon tumors after vehicle or MRTX1133 treatment. Scale bar: 500 μ m. Sections are from $n = 3$ mice per group and timepoint. **m–p**, Total number of macroscopic liver (**m**) and lung (**o**) metastases and representative H&E-stained sections of liver (**n**) and lung (**p**) metastases at the indicated timepoints. Sample sizes are as in **j**. Scale bars: 200 μ m (**n** and **p**). For **b–d**, **j**, **m** and **o**, each point represents a mouse. Data are presented as mean $\pm$ s.e.m. Groups were compared using the Mann–Whitney *U*-test (unpaired, no Gaussian distribution, two tailed). *P* values are shown in the corresponding panels. In vivo experiments were repeated twice. **D**, Day. Illustrations created in BioRender: **a**, Heinlein, M. <https://biorender.com/j33e534> (2025); **i**, Heinlein, M. <https://biorender.com/q84c829> (2025).

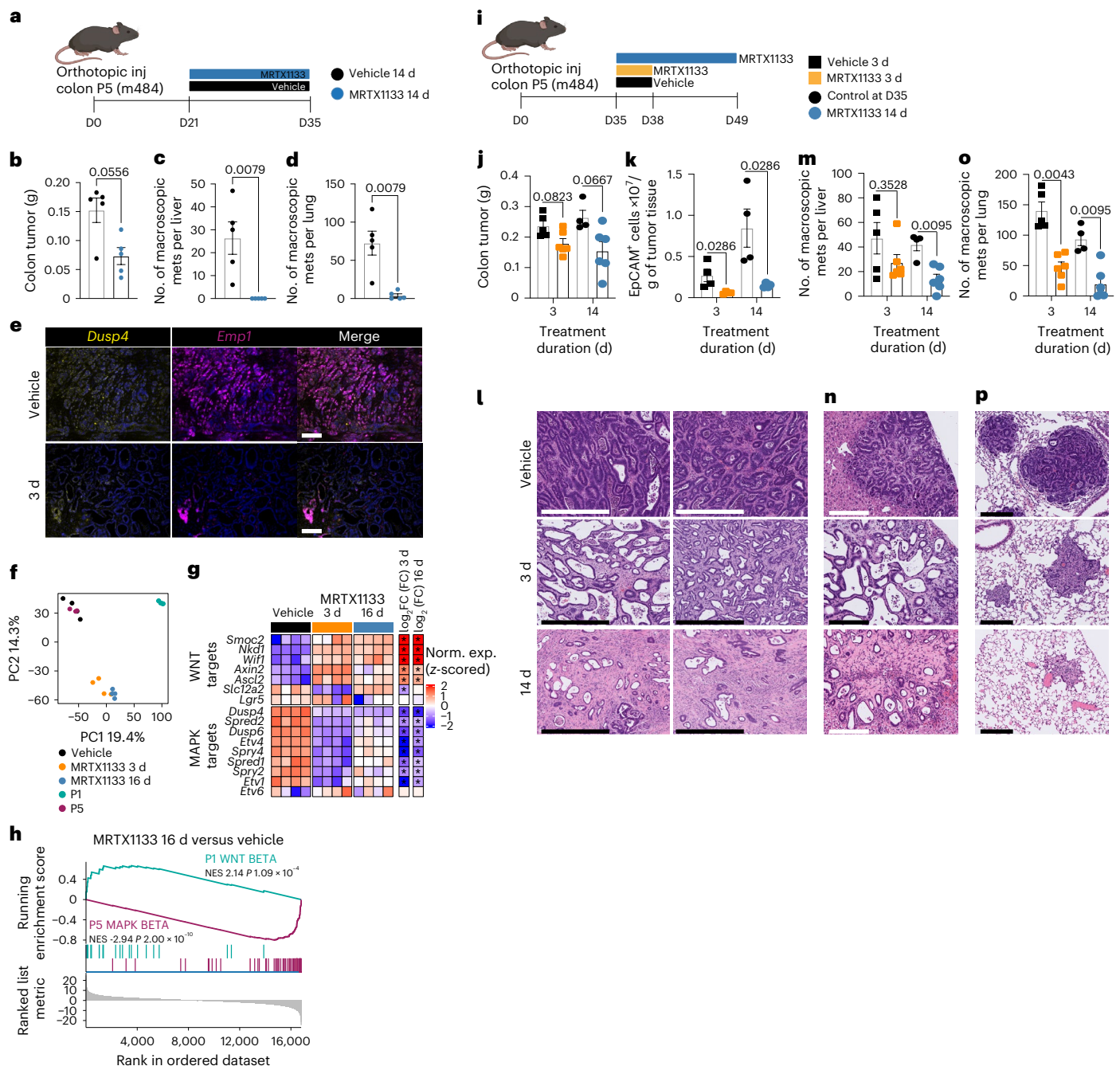
dataset (Extended Data Fig. 8c), KRAS mutation status alone did not significantly correlate with overall survival (Extended Data Fig. 8d), likely because MAPK pathway activation can arise through alternative mechanisms such as BRAF activation or KRAS amplification, which were not captured in this dataset. Nevertheless, this indicates that MAPK activation alone is insufficient as a prognostic marker and supports our approach of using combined MAPK and WNT pathway activity to provide improved prognostic resolution. Together, these findings suggest that the interplay between high MAPK activity and low WNT activity may serve as a potential biomarker for poor prognosis in CRC, underscoring the critical role of MAPK signaling in disease progression, metastasis and survival outcomes.

Discussion

Our study uncovered a dynamic interplay between the MAPK and WNT signaling pathways, where the level of MAPK activity appeared to be the limiting step for metastatic progression. We propose that MAPK

signaling induces a highly metastatic transcriptional state by suppressing a WNT-driven stem cell state in CRC. This inverse relationship aligns with observations in intestinal homeostasis, where MAPK and WNT signaling balance intestinal function^{29,30}. In CRC, this inverse relationship persists, with MAPK-high and WNT-high cell states being mutually exclusive^{31–33}. Tumor cell plasticity enables the transition between these states and facilitated adaptation during metastatic progression.

Both Lgr5⁺ and Lgr5[−] cells play distinct, yet essential, roles in tumor initiation and metastasis. Although Lgr5⁺ ISC are crucial during CRC initiation, they are dispensable for primary tumor growth. On depletion, these cells are constantly replenished by a highly proliferative Lgr5[−] population⁷. Consistent with our findings, this proliferative state, likely driven by MAPK signaling, downregulates WNT-driven stem cell genes. MAPK inhibition reverses this state, reactivating WNT signaling and Lgr5⁺ stem cell gene expression, consistent with the stem cell axis described in ref. 18, in which metastatic P5 cells predominantly occupy a regenerative RSC-like state but revert toward a CBC-like state on MAPK inhibition.



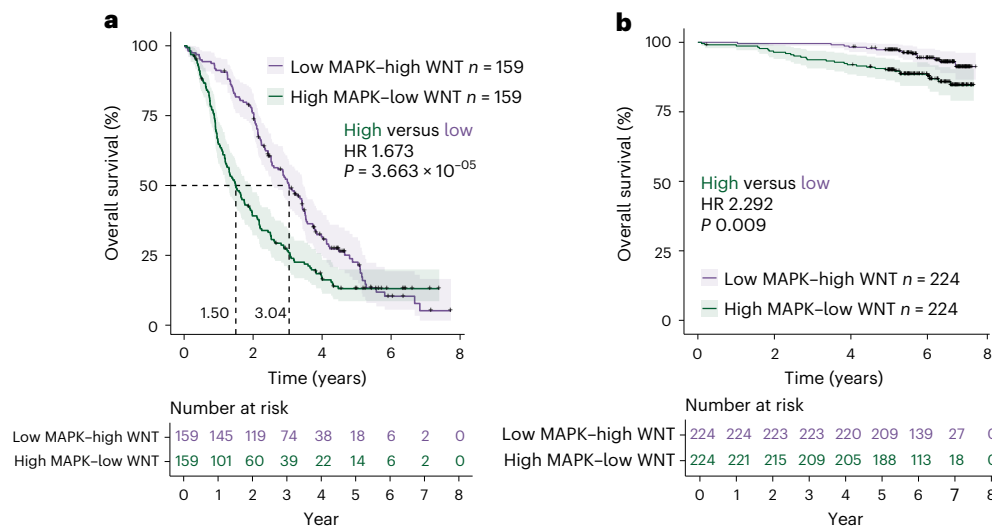


Fig. 6 | High-MAPK together with low-WNT activity is associated with shorter overall survival in patients with CRC. a, Kaplan–Meier survival curves showing overall survival for patients of the CALGB cohort with CRC for the MAPK target gene signature high–WNT target gene signature low group (high, dark green, $n = 159$ patients, median survival 1.50 years) and MAPK target gene signature low–WNT target gene signature high group (low, purple, $n = 159$ patients, median survival 3.04 years) ($P = 3.66 \times 10^{-5}$). **b**, Kaplan–Meier survival curves showing overall survival for patients of the AVANT cohort with CRC for the MAPK target

gene signature high–WNT target gene signature low group (high, dark green, $n = 224$ patients) and MAPK target gene signature low–WNT target gene signature high group (low, purple, $n = 224$ patients). In **a** and **b**, black crosses indicate censored patients. The solid lines represent Kaplan–Meier survival estimates, with shaded areas indicating 95% two-sided confidence intervals. Survival distributions were compared using the log(rank test). Hazard ratios (HRs) and P values were determined using Cox’s proportional hazards regression (two sided) without multiple-testing correction.

Metastasis-seeding cells typically lack *Lgr5* expression⁸ but can express markers such as *EMPI* (refs. 10, 34). Accordingly, we observed downregulation of WNT-driven stem cell genes in highly metastatic cells. In addition, our study connected increased MAPK signaling with high *Emp1* expression in metastatic tumors, thereby further expanding on prior findings that associate *EMPI*⁺ cells with increased migration, metastasis, advanced stages and relapse in patients with CRC^{10,24}. However, further mechanistic studies need to be conducted in the future to conclusively establish this relationship.

Disseminated CRC cells exhibit high plasticity, with *Lgr5*⁺ metastasis-seeding cells regaining *Lgr5* expression to initiate metastatic outgrowth^{8,10}. Notably, MAPK inhibition significantly reduced *Emp1* and increased *Smoc2* expression in colon tumors, curtailing liver metastasis formation when administered before metastatic outgrowth. Furthermore, we showed that high MAPK signaling drives the expression of a YAP gene signature in the metastatic cells, since KRAS inhibition reverses this signature. Qin et al. recently reported that APC loss and KRAS^{G12D} activation trap cells in a proliferative cancer stem cell state characterized by low YAP activity³⁵, suggesting mutual exclusivity between MAPK and YAP programs during CRC tumorigenesis. However, our findings of increased YAP activity in P5 metastatic cells despite elevated MAPK signaling suggest that the interaction between these pathways may shift in the metastatic context, which aligns with studies linking EGFR inhibition to reduced YAP-driven transcription, thereby favoring WNT-dependent states in residual tumor cells³⁶. YAP signaling has also been reported to restrict WNT activity and ISC expansion^{37,38}, whereas high MAPK signaling leads to stem cell loss³², further supporting our observations. The positive correlation between MAPK signaling and YAP transcriptional activity^{31,36} may contribute to a YAP-dependent state critical for metastatic colonization of the liver³⁹. These findings suggest that MAPK inhibition may disrupt both metastatic seeding and outgrowth by interfering with the highly metastatic cell state. However, the contribution of YAP and its transcriptional effector TEAD to the metastatic state and *EMPI* expression in CRC warrants further investigation.

Preclinical studies show varied anti-tumor responses to MRTX1133 in *Kras*^{G12D} mutant cancer xenograft models, with notably weaker

efficacy in CRC compared to pancreatic cancer⁴⁰. We show here that KRAS^{G12D} inhibition reduces metastatic burden both in the adjuvant setting and when used in mice with advanced metastatic disease. Notably, lung metastases showed greater sensitivity than liver metastases, likely reflecting site-specific microenvironmental cues such as growth factors that may modulate WNT^{41,42} and MAPK signaling⁴³ despite underlying mutations in these pathways. The contribution of these extracellular cues on the transcriptional cell state in different tissues and how KRAS^{G12D} inhibition in tumor cells affects the TME at both the primary and the metastatic sites, as well as the potential synergies with immune-checkpoint inhibitors, remain to be formally tested.

In conclusion, our study demonstrates that, although oncogenic signaling through both the WNT and the MAPK pathways is essential for CRC tumorigenesis, metastatic dissemination in CRC is driven by a MAPK-high and WNT-low transcriptional state. However, metastatic outgrowth requires a subsequent transition to a MAPK-low–WNT-high state indicating that, rather than one pathway being more important than the other, both are indispensable at distinct steps of the metastatic process, opening up opportunities for greater benefit when targeting these pathways in the adjuvant setting. In contrast, pharmacological inhibition of the high-MAPK signaling state using KRAS inhibitors in established tumors induces a WNT-high state, leading to residual disease and may help explain the limited efficacy of MAPK-targeted therapies in patients with CRC. We propose that this WNT reactivation constitutes a form of adaptive resistance, which will require the combination of agents targeting the MAPK and WNT pathways, an important area for future investigation as more selective downstream WNT pathway inhibitors become available. Another interesting avenue would be to explore epigenetic regulators that might prevent the switch from the MAPK-high to a WNT-high state after KRAS^{G12D} inhibition.

Methods

Ethical regulations

Our research complied with all relevant ethical regulations. All animal activities in this research study were conducted under protocols (nos. 19-1847, 19-1976, 24-0259 and 24-0694) approved by the Institutional

Animal Care and Use Committee and the Laboratory Animal Resource Committee at Genentech and adhered to the Guidelines for the Care and Use of Laboratory Animals. The maximal permitted tumor burden was defined as anal swelling with an ethical limit of 2.0 cm. The approved maximal tumor size and burden were not exceeded in any experiment. Additional humane endpoint criteria included >20% body weight loss, severe lethargy, hunched posture, decreased grooming, severe diarrhea, rectal prolapse or persistent bleeding. Animals were euthanized on reaching any of these criteria using CO₂ inhalation. For human data, all clinical trial participants provided preprocedural written informed consent for the collection and publication of molecular data.

Experimental animals

Wild-type female C57BL/6NJ (stock no. 005304) and female NSG (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ) (stock no. 005557) animals were purchased from the Jackson Laboratory. *VilCre^{ERT2}Apc^{fl/fl}Kras^{LSL-Q61R/+}* and *VilCre^{ERT2}Apc^{fl/fl}Kras^{LSL-G12D/+}* mice were bred at Genentech. Alleles used for this study were generated or licensed by Genentech from the appropriate institutions: *VillinCre^{ERT2}* (ref. 44), *Kras^{LSL-G12D/+}* (ref. 45), *Apc^{tm2Rak}* (designated *Apc^{fl/fl}*)⁴⁶ and *Kras^{LSL-Q16R}* (ref. 47). *Apc^{tm2Rak}* and *Kras^{LSL-Q16R}* were maintained on a C57BL/6N background. *Kras^{LSL-G12D/+}* mice were backcrossed on to C57BL/6N and verified by SNP genotyping. Crosses with *VillinCre^{ERT2}* resulted in a mixed C57BL/6N/J background. Mice were housed in specific pathogen-free facilities at Genentech in individually ventilated cages within animal rooms maintained at a constant temperature (20.0–26.1 °C) and humidity (30–70%) under a 14 h:10 h light-to-dark cycle and were allowed free access to standard diet (LabDiet 5010) and water. All experimental mice were females aged between 7 and 16 weeks.

Crypt isolation, organoid culture and genome editing to establish AKPS organoids

Wild-type mouse organoids were derived from isolated crypts of the small intestine of female C57BL/6N mice and maintained as previously described^{48,49}. Briefly, intestines were opened longitudinally, washed in ice-cold phosphate-buffered saline (PBS), cut into 0.5-cm to 1-cm fragments and incubated in 2.5 mM EDTA to release crypts. Crypt fractions were enriched by differential shaking, filtered through a 100-μm strainer and pelleted by centrifugation (150–200g, 5 min). Crypts were cultured in basal medium containing advanced Dulbecco's modified Eagle's medium (DMEM)/F12 (Gibco, cat. no. 126340-010), 2 mM GlutaMAX (Gibco, cat. no. 35050-061), penicillin–streptomycin (10,000 U ml^{−1}, Gibco, cat. no. 15140-122), 100 mM Hepes, pH 7.2 (Genentech), N2 (Gibco, cat. no. 17502-048), B27 (Gibco, cat. no. 17504-044), 1 mM N-acetyl-L-cysteine (Sigma-Aldrich, cat. no. A7250), 100 ng ml^{−1} of noggin (Peprotech, cat. no. 250-38), 200 ng ml^{−1} of R-spondin 3 (Genentech) and 50 ng ml^{−1} of EGF (R&D, cat. no. 2028-EG-200) in a 1:1 medium-to-Matrigel plug (Corning, cat. no. 356231).

AKPS organoids were generated from wild-type organoids by sequential CRISPR–Cas9 gene editing of *Apc*, *Kras*, *Trp53* and *Smad4* (Supplementary Table 8). The *Kras^{G12D}* mutation (exon 2) was introduced via homology-directed repair (Supplementary Table 8). Organoids were dissociated into single cells using accutase (Sigma-Aldrich, cat. no. A6964), followed by nucleofection with 150 pmol of guide RNA (IDT) and 60 pmol of Cas9 (V2, Invitrogen) in P1 primary cell solution (Lonza, cat. no. PBP1-00675) using a Lonza 4D nucleofector (program CA-137). Mutant organoids were selected as follows: withdrawal of R-spondin 3 (*Apc*), withdrawal of EGF (*Kras*), treatment with 5 μM Nutlin-3 (Sigma-Aldrich, cat. no. SML0580) (*Trp53*), withdrawal of noggin and treatment with 10 ng ml^{−1} of TGFβ1 (Novus Biologicals Inc., cat. no. 7666-MB-005) (*Smad4*).

To generate AKPS organoids from female *VilCre^{ERT2}Apc^{fl/fl}Kras^{LSL-Q61R/+}* and *VilCre^{ERT2}Apc^{fl/fl}Kras^{LSL-G12D/+}* mice, animals were treated with 100 mg kg^{−1} of tamoxifen (Sigma-Aldrich, cat. no. T5648) dissolved in

corn oil (Sigma-Aldrich, cat. no. C8267) by injection i.p. on 2 d consecutively. After 4 d, the small intestines were isolated, the organoids generated as described above and cultured in basal medium. Then, 1 week after seeding, R-spondin 3 and EGF were omitted from the medium to select for *Apc^{S80S}Kras^{G12D}* or *Apc^{S80S}Kras^{Q61R}* mutant organoids. Mutations in *Trp53* and *Smad4* were introduced and selected for as described above. These organoids are referred to as AK^{Q61R}PS and AK^{G12D}PS. The identity of the different organoid cells was authenticated by WES, RNA-seq or western blotting analysis. All cultures tested negative for *Mycoplasma* spp.

In vivo orthotopic passaging to generate metastatic organoids

The orthotopic injection method of organoids to establish primary colon tumors has been previously described⁷. Briefly, mice were anesthetized with isoflurane and injected subcutaneously with buprenorphine (Etiq XR 3.25 mg kg^{−1} every 72 h, Fidelis). Animals were then placed under a dissecting microscope (S4 stereomicroscope, Leica), under constant isoflurane inhalation. A blunt-ended hemostat (Halsted Mosquito Forceps, Roboz) was inserted 1 cm into the anus and used to transiently expose the mucosa. A 10-μl suspension of 1 × 10⁴ organoids in 100% Matrigel (Corning, cat. no. 356231) was injected into the colonic submucosa about 1 cm proximal to the anus (rectum). The hemostat was reinserted and the fold released, thereby reversing the rectal prolapse.

To generate an immunocompetent mouse model of mCRC, we employed in vivo orthotopic passaging. A total of 5 × 10⁴ AKPS parental organoids (P0) were injected orthotopically as described above. Animals were euthanized using CO₂ inhalation at the humane endpoint, defined in accordance with institutional animal ethics committee guidelines. The liver was then examined for macroscopic metastases using an S4 stereomicroscope (Leica) and the number of macro-metastases quantified. When present, primary tumors and liver metastases were harvested separately and enzymatically dissociated into single cells (0.1 mg ml^{−1} of DNase I (Roche, cat. no. 10104159001) and 0.1 mg ml^{−1} of Liberase TL (Roche, cat. no. 05401020001)). The resulting cells, independent of passage number, were cultured under identical conditions and seeded in a 50% Matrigel plug and cultured in AKPS organoid medium which enabled the selection and expansion of tumor cells only. After 7–10 d, 1 × 10⁴ organoids derived from liver metastases were orthotopically injected into a new cohort of C57BL/6N mice (Fig. 1c). This process was repeated for a total of five passaging rounds (Extended Data Fig. 1d). Survival data were analyzed using the standard Kaplan–Meier analysis and survival curves were compared using a log(rank test) (GraphPad Prism).

Hydrodynamic tail vein injections

To induce liver metastasis, we performed hydrodynamic tail vein injections⁵⁰. For this, a total of 1 × 10⁴ single-cell organoids were resuspended in 1.8 ml of PBS (Genentech) and injected intravenously in <8 s. Mice were euthanized 6–7 weeks post-injection. Liver metastases were quantified as described above.

Intrasplenic injections

Liver metastases were induced by intrasplenic injections as described previously⁵¹. Briefly, mice were anesthetized with isoflurane and injected subcutaneously with buprenorphine. After a subcostal incision, the spleen was exposed and a total of 5 × 10⁴ single-cell organoids resuspended in 50 μl of PBS were slowly injected into the spleen using a 27G needle. The spleen was removed by cautery and the incision site closed. Then, 4 weeks post-injection, mice were euthanized. Liver metastases were quantified as described above.

Drug treatments and cell viability assay

For in vitro experiments the small molecule inhibitor MRTX1133 (KRAS^{G12D} inhibitor, Chemgood or Genentech Compound Management),

EPZ004777 (DOT1L inhibitor, Selleck Chem, cat. no. S7353), RG108 (DNA methyltransferase inhibitor, STEMCELL Technologies, cat. no. 72212), A-485 (p300/CBP histone acetyltransferase inhibitor, STEMCELL Technologies, cat. no. 100-1645), panobinostat (broad-spectrum histone deacetylase inhibitor, Tocris, cat. no. 7629) and GNE-7883 (TEAD inhibitor, Genentech Compound Management) were dissolved in 100% dimethyl sulfoxide (DMSO) (Sigma-Aldrich), stored at -20°C and used at the indicated concentrations and durations. For western blotting analysis, organoids were grown for 72 h and treated with 100 nM MRTX1133 for 24 h. For real-time quantitative PCR (RT-qPCR) analysis organoids were seeded as single cells in 50% Matrigel, cultured for 24 h and treated with MRTX1133 alone for 24 h or in combination with TEAD for 48 h. To determine half-maximal inhibitory concentration values, organoids were dissociated into single cells using accutase. Single cells, 1×10^3 , were plated on a ultra-low attachment, round-bottomed, black-walled, 96-well plate (Corning). After 24 h, cells were treated with either DMSO or serial dilutions of MRTX1133 for 72 h. Cell viability was determined using the CellTiter-Glo 3D Cell Viability Assay (Promega, cat. no. G9681) according to the manufacturer's instructions and luminescence was measured on the Spectramax i3x (Molecular Devices, Softmax Pro v7). Dose-response curves were generated using GraphPad Prism. For the epigenetic inhibitor study, organoids were seeded in 50% Matrigel domes and treated for 8 d with EPZ004777 (10 μM), RG108 (10 μM), A-485 (10 μM) or panobinostat (50 nM) before being subject to ATAC-seq.

For in vivo drug treatment experiments, MRTX1133 was formulated in 10% Captisol (Selleck Chem, cat. no. S4592) in 50 mmol l^{-1} of citrate buffer, pH 5.0 (Teknova, cat. no. Q2443). The solution was stored at 4°C , protected from light and used within 1 week. MRTX1133 was administered via injection i.p. at a dose of 30 mg kg^{-1} , twice daily²⁵. Treatment start and duration are noted in the relevant figure legends.

Protein isolation and western blotting analysis

Organoids were harvested and cell pellets washed with PBS and snap frozen on dry ice. Lysates were generated using the Pierce radioimmunoprecipitation analysis lysis buffer (Thermo Fisher Scientific, cat. no. 89900) containing protease and phosphatase inhibitor (Roche, cat. nos. 05892791001 and 04906837001). Protein concentration was determined using the Pierce BCA Protein Assay kit (Thermo Fisher Scientific, cat. no. 23227) according to the manufacturer's instructions. Western blotting was performed according to standard protocols. After blocking in blocking buffer (Li-Cor, cat. no. 927-40000), membranes were incubated with the indicated primary antibodies at a 1:1,000 dilution: SMAD4 (D3R4N, Cell Signaling, cat. no. 46535), pMEK (41G9, Cell Signaling, cat. no. 9154), MEK (25/MEK1, BD, cat. no. 610122), pERK (D13.14.4E, Cell Signaling, cat. no. 4370), ERK (137F5, Cell Signaling, cat. no. 4695), pRSK (E238, Abcam, cat. no. ab32413), RSK (32D7, Cell Signaling, cat. no. 9355) and DUSP6 (EPRI29Y, Abcam, cat. no. ab22081), β -actin (8H10D10, Cell Signaling, cat. no. 3700) and β -tubulin (D3U1W, Cell Signaling, cat. no. 86298). The secondary antibodies were: IRDye 680LT goat anti-mouse immunoglobulin G or IRDye 800CW goat anti-rabbit immunoglobulin G (Li-Cor, cat. no. 926-68050 or 926-32211). The membranes were visualized on a Li-Cor Odyssey M Scanner.

RT-qPCR

Organoids were harvested and RNA isolated using the RNeasy Plus Mini kit or RNeasy Mini kit (QIAGEN, cat. no. 74134 or 74106) according to the manufacturer's instructions. After RNA isolation, complementary DNA was prepared using a High-Capacity cDNA Reverse Transcription kit (Thermo Fisher Scientific, cat. no. 4368814). Expression analysis was performed using TaqMan gene expression master mix (Thermo Fisher Scientific, cat. no. 4369016) on a Viia7 RT-PCR system (Thermo Fisher Scientific). Alternatively, RT-qPCR was performed using the TaqMan RNA-to-C_T-1-Step kit (Thermo Fisher Scientific, cat. no. 4392656) on a CFX BioRad Real-Time PCR Detection System. TaqMan gene expression assays (Thermo Fisher Scientific) were used for *Rpl19*-FAM (Mm02601633_g1),

Etv4-FAM (Mm00476696_m1), *Dusp6*-FAM (Mm00518185_m1), *Emp1*-FAM (Mm00515678_m1), *Dusp4*-FAM (Mm00723761_m1), *Lgr5*-FAM (Mm00438890_m1), *Smoc2*-FAM (Mm00491553_m1), *Nkd1*-FAM (Mm00471902_m1), *Ctgf*-FAM (Mm01192933_g1) and *Cyr61*-FAM (Mm00487498_m1). The cycle threshold (C_t) values from two technical replicates (for MRTX1133 single compound treatment) or three technical replicates (MRTX1133 and TEAD combination treatment) were assessed for each treatment concentration and condition. Relative gene expression levels were calculated using the comparative $\Delta\Delta\text{C}_t$ method, with *Rpl19* serving as the endogenous reference gene.

Tissue processing for histology and immunohistochemistry

The primary tumor was excised, weighed and its volume calculated ($0.523 \times \text{length} \times \text{width}^2$). Macroscopic liver and lung metastases were quantified as described above. Organs were fixed overnight in 10% neutral buffered formalin (the lungs flushed before fixation), washed in PBS, transferred to 70% ethanol for 24 h and paraffin embedded. Sections (4–6 μm) were stained with H&E using standard protocols and scanned on a Hamamatsu S360 microscope (NPD.view2).

Immunohistochemistry staining and imaging

Paraffin-embedded tissues were deparaffinized and rehydrated using standard protocols, then incubated in PBS containing 1% Tween-20 for 15 min. Antigen retrieval was performed in citrate buffer, pH 6.0, for 15 min using a pressure cooker. Endogenous peroxidase activity was quenched with 3% H_2O_2 in methanol for 30 min. Blocking was carried out using the Endogenous Blocking Kit (Invitrogen, cat. no. E21390) according to the manufacturer's instructions, followed by incubation with the M.O.M. Blocking Reagent (Vector Laboratories, cat. no. MKB-2213-1). Slides were incubated overnight at 4°C with mouse monoclonal anti- β -catenin antibody (15B8, Abcam, cat. no. ab6301). After washing, slides were incubated for 1 h at room temperature with a horseradish peroxidase-conjugated, anti-mouse secondary antibody. Signal was developed using the Tyramide Signal Amplification kit (Thermo Fisher Scientific), followed by counterstaining with DAPI and mounting in Pro-Long Gold Antifade Mountant (Thermo Fisher Scientific). Fluorescent images were acquired on a Leica DM4 B with a Leica DFC 7000 T camera (software Leica Application Suite X 3.4.2.18368).

RNAscope in situ hybridization

Hybridizations using the RNAscope method were performed according to the manufacturer's protocol (Advanced Cell Diagnostics) using the RNAscope 2.5 HD Reagent Kit-RED (cat. no. 322350) and the Multiplex Fluorescent detection kit (cat. no. 323110). Probes used were: Mm*Smoc2* (cat. no. 318541), Mm*Emp1* (cat. no. 446901) and C2 Mm*Dusp4* (cat. no. 425871). Fluorescent images were acquired on a Leica DM4 B with a Leica DFC 7000T camera. DAPI was used as a counterstain.

Tissue processing for cytokine analysis by Luminex assay

Tumor tissues were harvested, snap frozen and lysed by homogenizing 100 mg of tissue in 500 μl of tissue lysis buffer (Cell Signaling Technology, cat. no. 9803) supplemented with Complete Mini EDTA-free Protease Inhibitor Cocktail (Roche, cat. no. 11836170001). Homogenization was performed using a metal bead (QIAGEN, cat. no. 13117-500) on a TissueLyser II (QIAGEN). Lysates were incubated on ice for 10 min and clarified at 20,000g for 10 min at 4°C . Protein concentrations were determined with the Pierce Micro BCA Protein Assay Kit (Thermo Fisher Scientific, cat. no. 23235) and normalized to tissue weight. Cytokines and chemokines were quantified using the mouse cytokine Luminex assay (Millipore, cat. no. MCYTMG-70K-PX32) according to the manufacturer's instructions.

FACS staining, sorting and analysis

Organoids were dissociated with accutase. Tumor tissues were minced and enzymatically digested (Roswell Park Memorial Institute (RPMI)

1640 (Genentech) + 2% heat-inactivated fetal bovine serum (FBS) (R&D Systems, cat. no. S12410H), 0.8 mg ml⁻¹ of dispase (Life Tech. 17105041), 0.2 mg ml⁻¹ of collagenase P (Roche, cat. no. 11249002001) or 0.1 mg ml⁻¹ of Liberase TL (Roche, cat. no. 05401020001) and 0.1 mg ml⁻¹ of DNase I (Roche, cat. no. 10104159001) at 37 °C for 15 min, with gentle resuspension every 5 min using a wide-bore pipette. The supernatant was filtered (70-µm filter MACS Smart strainers, cat. no. 130-110-916) into RPMI medium + 10% heat-inactivated FBS and placed on ice. Digestion was repeated until a single-cell suspension was obtained. Red blood cells were lysed using ammonium–chloride–potassium (ACK; Genentech) buffer for 2–5 min at room temperature. Before cell surface staining, cells were incubated with Fc block (2.4G2; BD, cat. no. 553142, 1:500) for 5 min at 4 °C. Cells were labeled with the following anti-mouse antibodies (BioLegend; 1:100 unless otherwise stated) for 30 min at 4 °C: CD45 Brilliant Violet 605 (30-F11, cat. no. 103140), CD45 Alexa Fluor-700 (30-F11, cat. no. 103128), CD326 APC (G8.8, cat. no. 118214), CD31 phycoerythrin (PE)/cyanine 7 (390, cat. no. 102418), CD3 PE/cyanine 7 (17A2, cat. no. 100220), H-2Ld/H-2Db PE (28-14-8, cat. no. 114507, 1:200) and H-2Kb FITC (AF6-88.5, cat. no. 116506, 1:200). Live cells were identified by incubation with calcein blue (Invitrogen, cat. no. C1429, 1:500), Sytox Blue (Thermo Fisher Scientific, cat. no. S34857, 1:1,000) or propidium iodide (Life Technologies, cat. no. P3566, 1:10,000). Data were acquired on a Fortessa, Symphony (BD Biosciences) and analyzed using FlowJo (Tree Star, 10.9.0) or cells were sorted on a Fusion or Aria (BD Biosciences) for downstream nucleus or RNA extraction. Post-sort purity was >99%.

DNA isolation and WES

Organoids from different passages were expanded in vitro and DNA was extracted using the DNeasy Blood & Tissue Kit (QIAGEN, cat. no. 69506) according to the manufacturer's instructions. DNA quantity and quality were assessed using the Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific) and Genomic DNA ScreenTape on TapeStation 4200 (Agilent Technologies). Sequencing libraries were prepared from 200 ng of genomic DNA using the SureSelect XT HS2 DNA System (Agilent Technologies), quantified with the Qubit dsDNA HS Assay Kit and the average library size was determined using D1000 ScreenTape on TapeStation 4200. Libraries (750 ng) were hybridized to SureSelect XT Mouse All Exon capture probes (Agilent Technologies), requantified and resized as described above, pooled and sequenced on a NovaSeq 6000 (Illumina) to generate 50 M paired-end 75-bp reads per sample. Pre-processing was performed using GATK4 (ref. 52) against the GRCh38 mouse reference genome.

Identification of somatic variants from WES data

Somatic variants were identified from WES using the GATK4 'Functional Equivalence' workflow. Paired-end FASTQ files were quality checked (FastQC) and variant called using Mutect2 (as part of GATK4 (ref. 52)), LoFreq2 (ref. 53) and Strelka⁵⁴ with functional annotations provided by the Ensembl variant effect predictor⁵⁵. High-confidence variants identified in samples compared to the parental sample were selected based on variant type (missense, nonsense-mediated decay transcript, frameshift, start lost or stop gained) identification by ≥2 and an allele frequency >0.4.

Identification of CNVs from WES data

CNVs were identified using Control-FREEC⁵⁶ by providing bam files of the sample of interest and control (parental sample) with the following parameters: contaminationAdjustment = TRUE, minimalSubclonePresence = 30, noisyData = TRUE and sex = XX.

RNA isolation, bulk RNA-seq and analysis

RNA from in vitro grown organoids or sorted tumor epithelial cells from in vivo colon tumors was extracted using the RNeasy Plus Micro or Mini kit (QIAGEN, cat. no. 74034 or 74134) with on-column DNase

digestion (QIAGEN, cat. no. 12023460) according to the manufacturer's instructions. RNA quantity and quality were assessed using the Qubit RNA HS Assay Kit (Thermo Fisher Scientific) and RNA ScreenTape on TapeStation 4200. For low-input RNA-seq library generation, cDNA was generated from 2 ng of total RNA using Smart-Seq v4 Ultra Low Input RNA Kit (Takara); 150 pg of cDNA was used for sequencing library preparation with the Nextera XT DNA Sample Preparation Kit (Illumina). For samples with >100 ng of total RNA, libraries were prepared using the Truseq Stranded mRNA kit (Illumina). Libraries were quantified with Qubit dsDNA HS Assay Kit, sized using D1000 ScreenTape on TapeStation 4200, pooled and sequenced on NovaSeq 6000 (Illumina) (30 M single-end 50-bp reads per sample). RNA-seq data were analyzed using HTSeqGenie⁵⁷ in BioConductor⁵⁸: low-quality reads (70% of bases with quality <23) or matches to ribosomal RNA and adapter sequences were removed; the remaining reads were aligned to the mouse reference genome (GRCh38.p5) using GSNAP^{59,60} (parameters: '-M 2 -n 10 -B 2 -i 1 -N 1 -w 200000 -E 1 --pairmax-rna=200000 --clip-overlap') and quantified by counting reads mapping against exons of each gene from the GENCODE M15 annotation.

Identification of differentially expressed genes and gene set analysis

Counts were normalized by the trimmed mean of M values (edgeR⁶¹) and differential expression assessed using Limma voom⁶² in R. Gene set enrichment analysis (GSEA) was conducted utilizing the fgsea⁶³ package, which assesses differences in ranking both within and outside of specific gene sets. In addition, over-representation analysis was performed with clusterProfiler⁶⁴, via a hypergeometric test to determine whether the identified genes were over-represented in the gene sets of interest. Multiple-testing adjustments were performed for differential expression analysis and GSEA. The Benjamini–Hochberg method was used to control the FDR^{62,63}. For differential expression analysis, effect sizes for comparison between groups A and B are log₂(fold-change) of normalized expression values between groups, that is: (Mean (log₂(normalized expression)) in group A) – (Mean (log₂(normalized expression)) in group B). For GSEA, effect sizes are normalized enrichment scores (https://docs.gsea-msigdb.org/#GSEA/GSEA_User_Guide).

Batch effect correction

RNA-seq data derived from the P1 (m4) and P5 (m484), in vivo sorted tumor epithelial cells and RNA-seq data from mice with colon tumors treated with MRTX1133 were combined and the batch effect adjusted using the comBat⁶⁵ R package utilizing the P5 samples in the former and the vehicle samples in the latter as anchor.

scRNA-seq library preparation and analysis

NSG mice were implanted orthotopically with 5 × 10⁴ AKVPSL organoids⁷ and developed primary colon tumors as well as liver and lung metastases. At the humane endpoint, liver and lung metastases were isolated, digested as described above and live RFP⁺ tumor epithelial cells sorted from each metastatic site and subjected to scRNA-seq, profiling ~10,000 cells per tissue.

Sample processing for scRNA-seq was done using Chromium Single Cell 3' Library reagent kit v2 following the manufacturer's guide (10x Genomics). Cell density and viability were determined using the Vi-CELL XR cell counter (Beckman Coulter) or manual hemocytometer counting. Approximately 10,000 cells were loaded on each chip. The cDNAs and libraries were prepared following the manufacturer's instructions (10x Genomics). Libraries were profiled by TapeStation D1000 kit and quantified using 1Qubit HS dsDNA kit (Invitrogen). Each library was sequenced in one lane of HiSeq4000 (Illumina) following the manufacturer's sequencing specification (10x Genomics).

Sample demultiplexing, barcode processing and single-cell 3'-gene counting were performed using 10X Cell Ranger. Reads were aligned to GRCh38 mouse reference assembly. Low-quality cells with >25%

mitochondrial reads or <200 detected genes were excluded. Per-gene unique molecular identifier counts were normalized by the total number of transcripts per cell, scaled by the median number of transcripts across all cells and log(transformed) for downstream analysis.

Dataset cell clustering and visualization were performed following the OSCA guidelines (<https://osca.bioconductor.org>)⁶⁶. Dimensionality reduction and clustering were performed to obtain a common Uniform Manifold Approximation and Projection coordinate space and grouping for all cells. Clustering and visualization were based on the top 20 PCs. We used the 50 nearest neighbors ($k = 50$) and a resolution of 0.3 for graph-based clustering. Gene signatures were scored using the UCell R package⁶⁷.

Bulk ATAC-seq and processing

ATAC-seq libraries from in vivo sorted P1 and P5 tumor epithelial cells were prepared as previously described⁶⁸. Briefly, sorted cells were spun at 500g for 5 min at 4 °C, lysed in ice-cold ATAC resuspension buffer [10 mM Tris-HCl, pH 7.4, 10 mM NaCl and 3 mM MgCl₂] containing 0.1 % NP-40 (Sigma-Aldrich), 0.1% Tween-20 (Sigma-Aldrich) and 0.01% digitonin (Promega), pipetted up and down 3× and incubated on ice for 3 min. Cells were then washed in ATAC resuspension buffer containing 0.1% Tween-20. Cell pellets were resuspended and DNA was tagmented using the enzyme and buffer provided in the Nextera Library Prep Kit (Illumina, cat. no. FC-121-1031). Tagmented DNA was then purified using the MinElute PCR purification kit (QIAGEN, cat. no. 28004). Libraries were amplified with NEBNext High-Fidelity 2x PCR Master Mix (NEB, cat. no. M0541S) and primers (universal primer 1 AATGATACGCGACACCGAGATCTACACTCGTCGGCAGCGTCAGATGTG and primer 2 with TruSeq index CAAGCAGAAGACGGCATACGAGAT[index] GTCTCGTGGGCTCGGAGATGT). Amplified libraries were cleaned up using QIAGEN MinElute Cleanup kits (cat. no. 28004). Libraries were quantified using the Qubit dsDNA HS Assay Kit and profiled using the D5000 ScreenTape on TapeStation 4200. Libraries were pooled and sequenced on an Illumina HiSeq 2500 sequencer to generate 25 M paired-end 50-bp reads per library.

Organoids treated with epigenetic inhibitors were harvested on ice and resuspended in 300 µl of ice-cold ATAC lysis buffer (Active Motif, cat. no. 53150). Nuclei were released by pipetting and incubation on ice for 2 min. A target of 100,000 nuclei per sample was processed for bulk ATAC-seq library preparation using the ATAC-Seq Kit (Active Motif) according to the manufacturer's instructions. ATAC-seq libraries were quantified using Qubit dsDNA Quantification kit and High Sensitivity D1000 ScreenTape (Agilent). Sequencing was performed on an Illumina NextSeq2000 platform, generating 35 M reads per sample.

ATAC-seq data were processed using the ENCODE ATAC-seq pipeline⁶⁹. In brief, adaptors were trimmed using cutadapt⁷⁰. ATAC-seq reads were aligned to the mouse genome (mm10) using Bowtie2 (ref. 71) and filtered for quality and duplicates using samtools⁷² and Picard (<https://broadinstitute.github.io/picard/>). Peaks were called using MACS2 (ref. 73). Specifically, 5'-ends of reads were shifted in a strand-specific manner to account for the Tn5 shift and identify the precise cut sites. The shifted read-start coverage was aggregated over both strands and smoothed using a 150-bp window for peak calling in MACS2. The called peaks were filtered using exclusion lists that contain genomic regions resulting in an anomalous, unstructured or experiment-independent high signal⁷⁴. The irreproducible discovery rate⁷⁵ framework was used to estimate reproducibility and stringent peak calls. To assess data quality, we measured read mapping statistics, enrichment quality control metrics, library complexity and reproducibility ratio and focused on the transcription start site enrichment score and the total number of peaks identified.

Bulk ATAC-seq analysis

The DiffBind package⁷⁶ was used to process and analyze the ATAC-seq data from both experiments. In short, bam files and narrowPeak files

were used as input to the dba.count function to count reads in binding site intervals. Counts were then normalized using Deseq2 based trimmed mean of M-values. PC analysis was then conducted on the normalized binding matrix using the R function prcomp. Differential accessible regions between P1 and P5 organoids or tumor epithelial cells were defined as log₂(fold-change) > 1.5 and FDR < 0.0001.

Motif enrichment analysis

HOMER⁷⁷ was used for motif discovery in open chromatin regions in P1 (m4) and P5 (m484) colon epithelial cells against a background of non-differentially accessible chromatin regions using the following command: findMotifsGenome.pl dar.bed -bg bg_peak.bed -size 200 -mask.

To evaluate whether there was a promoter or enhancer bias in the enrichment results, we performed the same HOMER analysis restricting on either promoter regions (annotated using the package Genomic Ranges⁷⁸ promoter method with the GRCm38 database and default parameters) or enhancer regions (the rest).

Direct target gene inference

BETA²³ was conducted to integrate differential accessible regions with differentially expressed genes, achieving the inference of direct target genes. Using the top enriched motifs identified by HOMER in the motif enrichment analysis, we identified DAR peaks that contain these motifs through the HOMER command findMotifsGenome.pl. We then provided these motif-harboring DAR peaks along with the gene fold-changes from the differential expression analysis as inputs to the BETA program. The BETA command was as follows: BETA basic -p [peak file] -e [expression file] -k BSF --info 1,2,3 -g mm10 --gname2 --da 2000 -d 100000 -c 0.1 -o [output].

In the output, inferred gene targets were ranked by the rank product metric. We focused on the top enriched motifs on both P5 peaks (AP-1 7-bp sequence (TGA[C][A]TCA)—BATF, ATF3, FRA1, JUNB, AP-1, FOS and BACH2) and P1 peaks (CREB 8-bp sequence (TGA[CG]TCA)—ATF1, ATF2, ATF7 and CREB5—and Wnt (CCTTTGA)—LEF1, TCF3, TCF7 and TCF12) and filtered on highly confident predicted targets (rank product < 0.0001). The P5 MAPK signature was based on P5 genes predicted to be exclusively regulated by AP-1 TF targets. The P1 WNT signature was based on P1 genes predicted to be regulated by WNT TF targets exclusively or in conjunction with the AP-1 and CREB TF targets.

Human datasets

AVANT. AVANT²⁷ cohort data were collected from a phase 3 randomized controlled clinical trial testing the efficacy of bevacizumab + oxaliplatin-based chemotherapy as adjuvant treatment for colon cancer. RNA was extracted from tumors of treatment-naïve patients before initiation of adjuvant therapy. Gene signature scores were calculated based on log(count per million) values of the RNA-seq data using R package multiGSEA (<https://github.com/lianos/multiGSEA/>) (function eigenWeightedMean). The MAPK signature score was calculated for each patient based on canonical MAPK downstream target genes (*SPRED1*, *SPRED2*, *ETS*, *SPRY2*, *SPRY4*, *ETV1*, *ETV4*, *ETV6*, *DUSP4* and *DUSP6*). The WNT signature score was calculated for each patient using key WNT downstream genes (*LGR5*, *AXIN2*, *ASCL2*, *SLC12A2*, *GKN33P*, *NKD1*, *WIF1* and *SMOC2*). Patients were binarized to high or low groups according to median scores for each signature separately. Patients falling into the high-MAPK and low-WNT category or low-MAPK and high-WNT category were presented in the Kaplan–Meier plots and 826 patients with RNA-seq data were used in this analysis, of whom 335 were women and 491 men.

CALGB. CALGB²⁸ cohort data were collected from a randomized phase 3 trial to determine drug combination efficacy in previously untreated patients. Gene signature scores were derived in the same way as in the AVANT cohort and 572 patients with RNA-seq data were used in this analysis.

Survival analysis

The R package *survival*⁷⁹ was used to perform survival analysis. The Kaplan–Meier curve of overall survival was used to compare the prognosis across subtypes (function *survfit*). The standard Cox's proportional hazard modeling (function *coxph*) was applied to calculate the HR between patients with high-MAPK plus low-WNT signature scores and low-MAPK plus high-WNT signature scores. The same type of model was also built to compare between patients who were KRAS wild-type and those who were KRAS mutant in the AVANT cohort. The log(rank test) was used to test the differential survival outcomes between groups.

Statistical analysis and reproducibility

No statistical methods were used to predetermine sample sizes but were informed according to previous experience with similar experiments and previous publications⁷. Statistical analyses were performed using Graph-Pad Prism (v10.4.0). For in vivo inhibitor studies, mice with colon tumors were randomized to ensure comparable tumor burden as assessed by anal swelling and body weight across treatment groups. Mice that did not develop tumors (for example, due to failed injection), required euthanasia before the predefined endpoint or developed nonphysiological tumors extending into the peritoneal cavity were excluded from downstream analyses. No data were excluded from in vitro experiments.

Normality and variance were not formally tested. For in vivo experiments and other datasets where a Gaussian distribution could not be assumed, nonparametric two-tailed Mann–Whitney *U*-tests were used unless otherwise stated. For in vitro experiments, data distribution was assumed to be normal but was not formally tested and two-tailed, unpaired Student's *t*-tests were applied. Data are presented as the mean ± s.e.m. unless otherwise indicated and individual data points are shown to illustrate data distribution. A *P* ≤ 0.05 was considered statistically significant. Exact *P* values are reported in the figures. Experiments were repeated independently where feasible, as specified in the figure legends. In vivo experiments comparing P1 and P5, including drug treatment studies, were repeated at least twice unless otherwise stated. Sequencing experiments were performed once with at least three biological replicates, with the exception of WES for which only one replicate per organoid line was assessed. Data collection and analysis were not performed blind to the conditions of the experiments.

Materials availability

Requests for mouse organoid models and other reagents generated in this study will be available upon reasonable request. These materials will be distributed by Genentech upon completion of a materials transfer agreement.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Mouse bulk RNA-seq, scRNA-seq and ATAC-seq data that support the findings of this study have been deposited in the Gene Expression Omnibus under SuperSeries accession no. [GSE283937](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE283937). Mouse WES data have been deposited in the Sequence Read Archive with accession nos. [PRJNA1196349](https://www.ncbi.nlm.nih.gov/sra/PRJNA1196349) and [SRP559178](https://www.ncbi.nlm.nih.gov/sra/SRP559178). Human RNA-seq data from the AVANT²⁷ and CALGB²⁸ studies are available in the European Genome-phenome Archive under study no. [EGAS00000001231](https://www.ebi.ac.uk/ena/browser/study/EGAS00000001231). All other data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided with this paper.

Code availability

No customized code was generated for this study. R code with intermediate data was deposited on GitHub (https://github.com/Genentech/2025_heinlein_et al) and is publicly available via Zenodo at <https://doi.org/10.5281/zenodo.17993514> (ref. 80).

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Author contributions

F.J.d.S. supervised the study. M.H. and F.J.d.S. designed and planned the study. M.H. engineered and characterized the different organoid lines, performed in vitro organoid experiments, conducted flow cytometry experiments and isolated samples for sequencing. M.H. and N.K. performed in vivo studies. A.R.M performed western blotting analysis and epigenetic inhibitor and TEAD inhibitor studies.

B.B. performed RNAscope and RT-qPCR analysis. S.B. performed scRNA-seq studies. M.S and Y.L. contributed next-generation sequencing data. G.X.L., L.T., F.M. and R.P. analyzed and helped interpret bioinformatics data. H.K. performed histopathological evaluation of colon tumors, liver and lung tissues. N.R. helped discuss the results. M.H. and F.J.d.S. wrote the paper. All authors contributed to discussing the results and commenting on the manuscript.

Competing interests

All authors are or were employed by Genentech at the time of their contributions to the work and own Roche shares.

Additional information

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Correspondence and requests for materials should be addressed to Robert Piskol or Frederic J. de Sauvage.

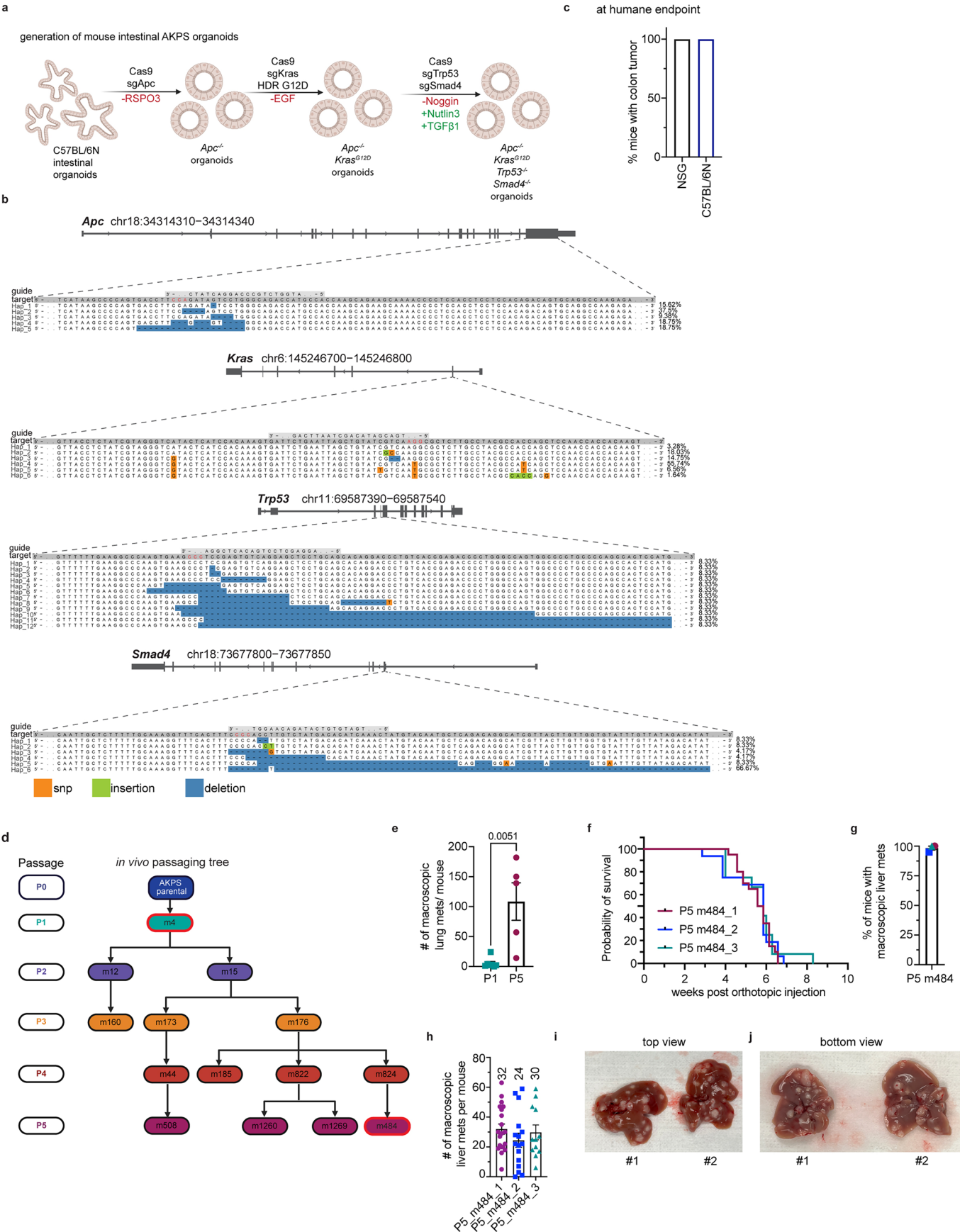
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Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Generation of a reliable immunocompetent mouse model of metastatic CRC. a, Schematic of the stepwise generation of *Apc*^{-/-}, *Kras*^{G12D}, *Trp53*^{-/-}, *Smad4*^{-/-} (AKPS) small intestinal organoids from C57BL/6 N mice using CRISPR/Cas9 including the selection media used to select for individual mutations. Illustrations in **a** created in BioRender; Heinlein, M. <https://biorender.com/y25q621> (2025). **b**, Haplotypes of *Apc*, *Kras*, *Trp53* and *Smad4* of the parental AKPS (P0) organoids. Only the regions harboring the binding sites of the guide RNA sequences are shown. Snp - single nucleotide polymorphism; Chr - chromosome.

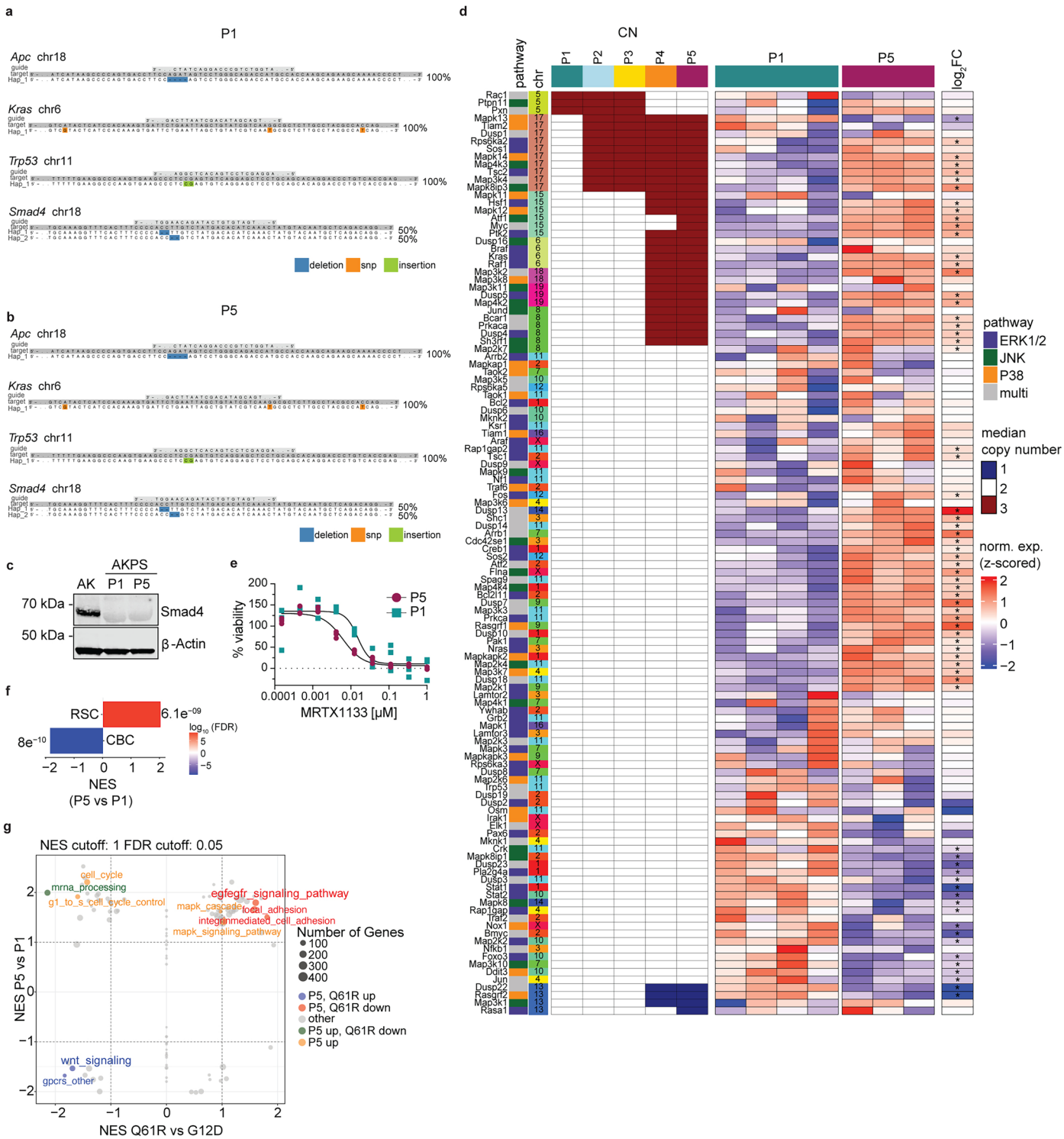
c, Percentage of NSG mice ($n = 10$) and C57BL/6 N mice ($n = 13$) presenting with a primary colon tumor at humane endpoint following orthotopic injection of parental (P0) AKPS organoids. **d**, Schematic of the in vivo orthotopic passaging strategy used to enhance metastatic potential of AKPS organoids. Numbers indicate animal identifiers from which metastases were isolated, expanded in vitro, and reinjected into new cohorts of C57BL/6 N mice. Passage 1 (P1; m4) and Passage 5 (P5; m484) lines used for subsequent experiments are highlighted in red. Schematic in **d** created in BioRender; Heinlein, M. <https://biorender.com/v45k881> (2025). **e**, Total number of macroscopic lung metastases per C57BL/6 N

mouse at humane endpoint after orthotopic injection of AKPS P1 (m4) or P5 (m484) organoids into the colon. P1 (m4) $n = 7$, P5 (m484) $n = 5$ mice. Groups were compared with Mann-Whitney *U* test (unpaired, no Gaussian distribution, two-tailed). *P*-values are shown in the corresponding panel. **f-h**, Survival plots (**f**), percentage of C57BL/6 N mice presenting with macroscopic liver metastases (**g**) and total number of macroscopic liver metastases per C57BL/6 N mouse (**h**) at humane endpoint after orthotopic injection of AKPS P5 (m484) organoids. Data presented is collated from three independent experiments. Repeat 1 $n = 20$ mice; Repeat 2 $n = 16$ mice; Repeat 3 $n = 12$ mice. For **g**, each dot represents the percentage of mice with metastases for each repeat. Note - data presented for P5_m484_1 is the same as presented in Fig. 1d. For **h**, each dot represents an individual mouse. The number above each bar graph shows the mean number of metastases per mouse. Note - data presented for P5_m484_1 is the same as presented in Fig. 1e. **i-j**, Representative images of livers from C57BL/6 N mice following orthotopic injection of AKPS P5 (m484) organoids at humane endpoint, shown from top (**i**) and bottom (**j**) views. **e, g, h**, Data are presented as mean $\pm$ s.e.m.



Extended Data Fig. 2 | Tumor-intrinsic changes enhance metastatic potential in vivo passaged organoids. **a**, Total number of somatic mutations (missense, nonsense-mediated decay (NMD), frameshift, start lost or stop gained) relative to the parental P0 sample in P1 (m4), P2 (m15), P3 (m176), P4 (m484) and P5 (m484) organoids, identified by at least two of three variant callers with allele frequency >0.4. NMD, nonsense-mediated decay. **b**, Somatic mutations in individual genes in P1 (m4), P2 (m15), P3 (m176), P4 (m484) and P5 (m484) organoids. NMD, nonsense-mediated decay. **c**, Genome-wide copy number alterations in P2 (m15), P3 (m176), P4 (m484) organoids across the whole mouse genome. red, copy number gain; blue, copy number loss; chr #, chromosome number. **a–c**, $n = 1$ replicate per organoid line. **d**, Gating strategy for FACS sorting of live EpCAM⁺ tumor epithelial cells from colon tumors for downstream sequencing experiments. Numbers indicate the percentage of cells within each gate. **e**, Normalized gene set enrichment score of all Wikipath gene signatures significantly enriched in P5 (m484) (red) or P1 (m4) tumor epithelial cells (blue). Color gradients represent statistical significance. NES, normalized enrichment score. **f**, Doubling time of P1 (m4) (teal) and P5 (m484) (maroon) organoids in vitro. Data pooled from two independent biological experiments (average of $n = 4$ technical replicates per experiment) and presented as mean. **g**, Primary colon tumor weight (circles) with corresponding number of macroscopic liver metastasis (squares) in mice orthotopically injected with P1 (m4) (teal) and P5 (m484) (maroon) organoids into the colon at humane endpoint. $n = 7$ mice per group. **h**, Total number of macroscopic liver (left) and lung (right) metastases per mouse following hydrodynamic tail vein (HTV) injection of P1 (m4) or P5 (m484) organoids. Liver: P1 (m4) $n = 10$; P5 (m484) $n = 10$. Data representative of two independent experiments. Lung: P1 (m4) $n = 9$; P5 (m484) $n = 8$ mice. **i**, Total number of macroscopic liver metastases per mouse following intrasplenic

injection of P1 (m4) ($n = 4$) or P5 (m484) ($n = 5$) organoids. **j**, Interferon-gamma (IFN γ) levels in pg/mL in the TME of P1 (m4) and P5 (m484) colorectal tumors harvested 4 weeks after orthotopic injection, as measured by Luminex assay, with $n = 5$ mice per group. **k**, Bulk RNA-sequencing expression (logCPM) of *lfngr1* and *lfngr2* in tumor epithelial cells isolated from P1 (m4) ($n = 4$) and P5 (m484) ($n = 3$) colorectal tumors. Differential expression was assessed using limma-voom. Box plots show the median (center line), interquartile range (Q1–Q3) and 1.5 $\times$ interquartile range whiskers. **l**, CXCL9 and CXCL10 in pg/mL in the TME of P1 (m4) and P5 (m484) colorectal tumors harvested 4 weeks after orthotopic injection, as measured by Luminex assay with $n = 5$ mice per group. **m**, Percentage CD3⁺ of CD45⁺ cells in the TME of P1 (m4) ($n = 10$) and P5 (m484) ($n = 8$) colorectal tumors 5–6 weeks post orthotopic injection as determined by FACS analysis. **n**, Representative flow cytometry histograms (left) and quantification of geometric mean fluorescence intensity (gMFI; right) showing surface expression of H-2K^b (top) and H-2D^b (bottom) in P1 (m4) and P5 (m484) organoids (one technical replicate each), with splenocytes included as a positive control. Fluorescence minus one (FMO) controls are shown for spleen, P1, and P5 samples. Vertical lines indicate median fluorescence values for the P1 and P5 FMO samples. Data representative of 2 independent experiments. **o**, Total number of macroscopic liver metastases per NSG mouse at humane endpoint following orthotopic injection of the indicated organoid lines. P0 $n = 10$; P2 (m15) $n = 18$; P5 (m484) $n = 8$ mice. Data for P0 is the same as a presented in Fig. 1b. **h–j**, **l**, **m**, **o** n represents the number of individual mice per group. Data are from a single independent experiment and presented as mean $\pm$ s.e.m. Groups were compared with Mann-Whitney U test (unpaired, no Gaussian distribution, two-tailed). P -values are shown in the corresponding panels.

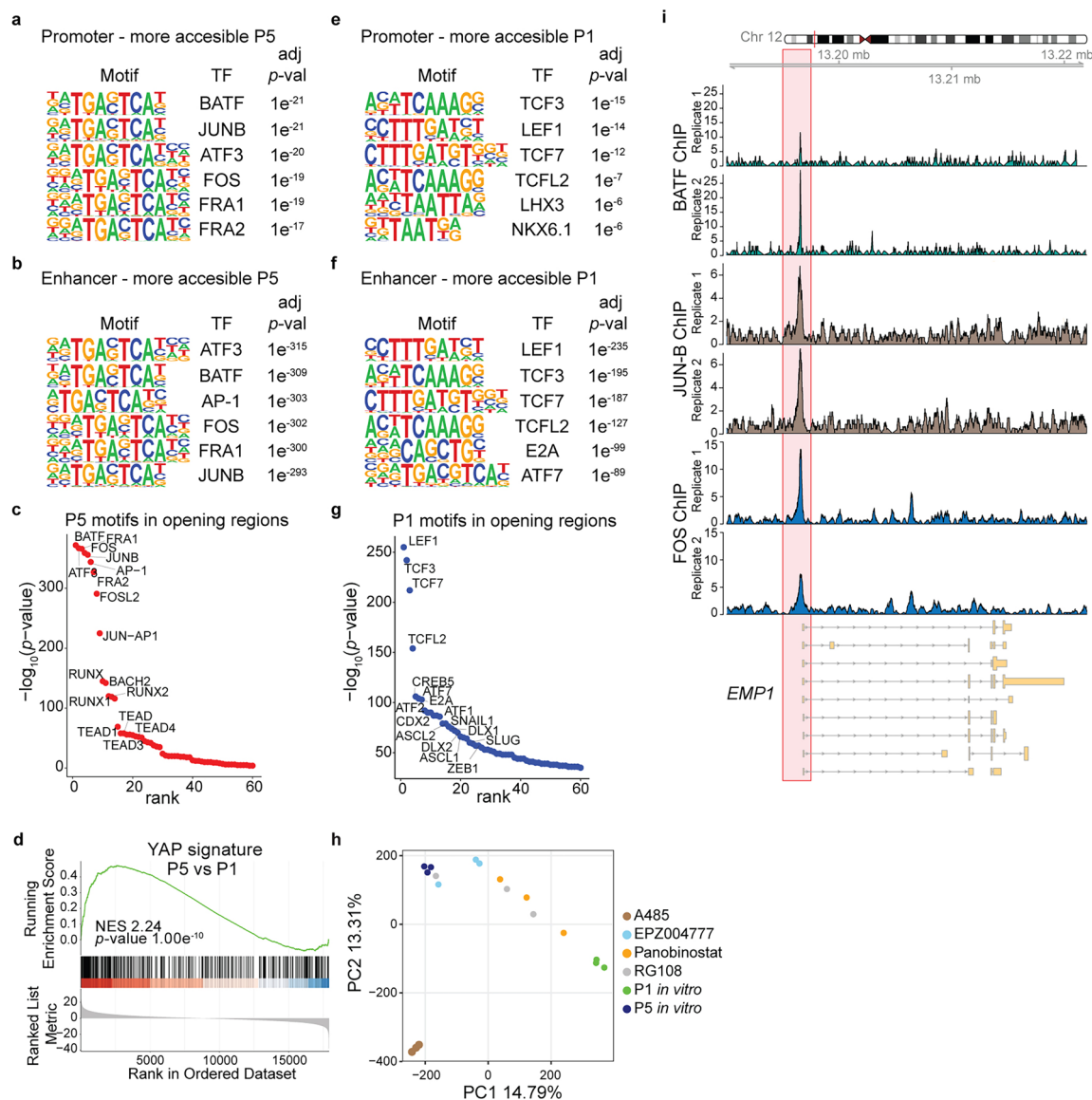


Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | P5 organoids show increased expression of MAPK-associated genes and less sensitivity to MAPK pathway inhibitors.

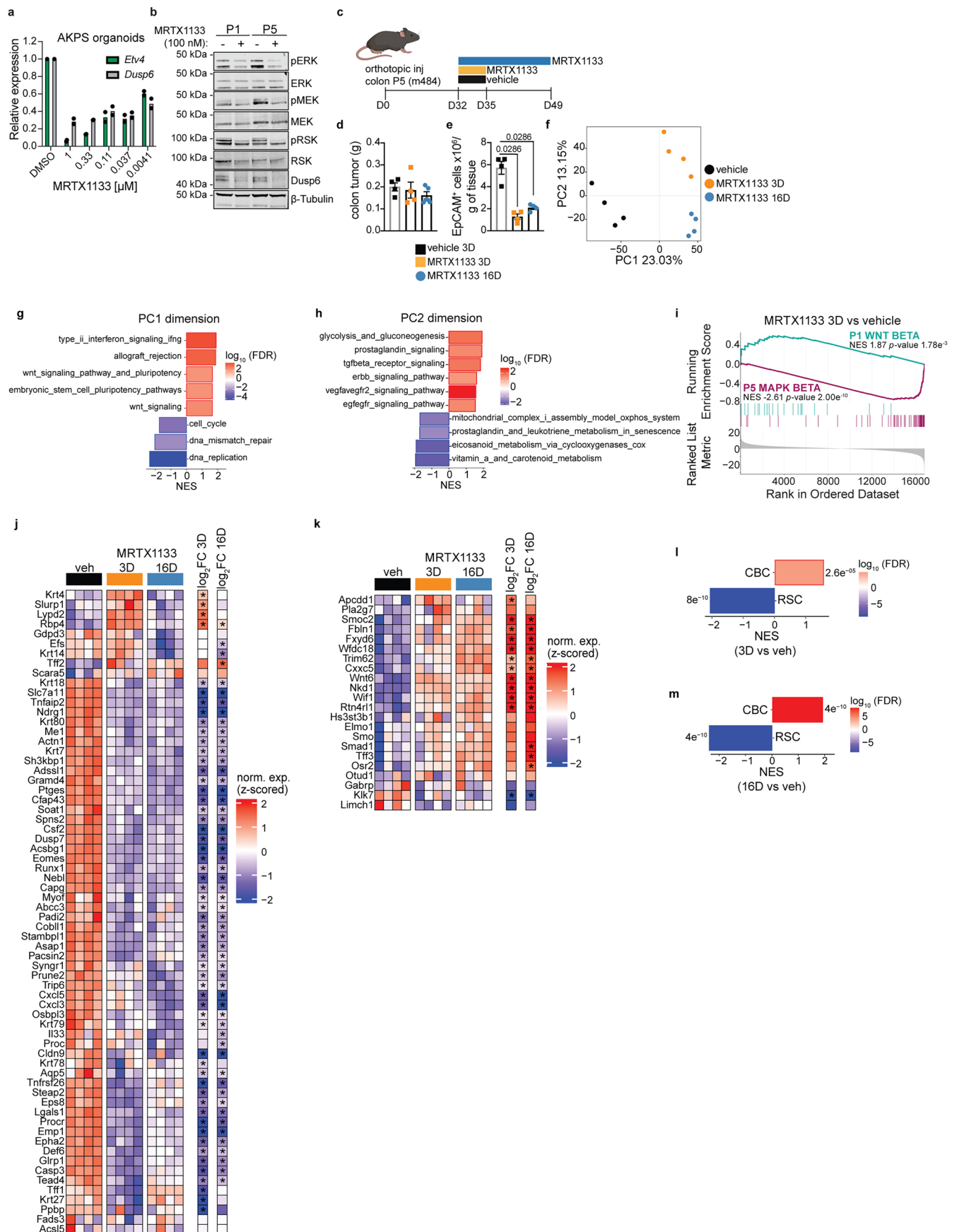
a – b, Haplotypes of *Apc*, *Kras*, *Trp53* and *Smad4* in P1 (m4) (**a**) and P5 (m484) (**b**) organoids. Only genomic regions encompassing guide RNA binding sites are shown. SNP, single nucleotide polymorphism. **c**, Western Blot analysis of Smad4 in AK, P1 (m4) and P5 (m484) organoids. Data from a single experiment. **d**, Copy number changes of genes within MAPK related pathways¹⁷ in P1 (m4), P2 (m15), P3 (m176), P4 (m824) and P5 (m484) organoids relative to P0 (parental) organoids ($n = 1$ replicate per organoid line) together with normalized expression (z-scored) and log₂ fold change of the same genes in tumor epithelial cells sorted from P1 (m4) ($n = 4$) and P5 (m484) ($n = 3$) colon tumors, 5 weeks post orthotopic injection. Asterisks indicate significant fold changes ($FDR \leq 0.05$). The exact FDR values for all fold changes are provided in Supplementary Table 1. Chr, chromosome; CN, copy number; stand.exp., standard expression; FDR, false

discovery rate. **e**, Dose–response curves for MRTX1133 in P1 (m4) and P5 (m484) organoids assessed 72 h after treatment. Data are from a single experiment and are presented as mean from $n = 3$ technical replicates per condition. **f**, Normalized gene set enrichment score for the CBC and RSC gene signatures¹⁸ enriched in P5 (m484) tumor epithelial cells (red) or P1 (m4) tumor epithelial cells (blue). Numbers indicate statistical significance. NES, normalized enrichment score. **g**, Two-dimensional plot showing normalized enrichment scores (NES) for pathways in sorted tumor epithelial cells from P5 (m484) ($n = 3$ mice) vs P1 (m4) ($n = 4$ mice) (y-axis) and AK^{Q61R}PS vs AK^{G12D}PS organoid cells ($n = 4$ replicates each) (x-axis). Each point represents a pathway, with size proportional to the number of genes in the pathway. Key pathways enriched are labeled and colors denoting regulation patterns: blue, downregulated in both comparisons; red, upregulated in both; green, up in P5 only and down in AK^{Q61R}PS; orange, up in P5 only. NES cutoff = 1; FDR cutoff = 0.05.



Extended Data Fig. 4 | Differential chromatin accessibility in P5 and P1 tumor cells. **a - b**, Logo plots show the six most enriched transcription factor motifs and associated transcription factors in promoter (**a**) and enhancer (**b**) regions that open in P5 (m484) tumor epithelial cells. Adjusted p -values derived from one-sided hypergeometric tests with Benjamini-Hochberg false discovery rate (FDR). **c**, Top 60 transcription factor motifs enriched in regions with increased accessibility in P5 (m484) compared with P1 (m4) colon tumor epithelial cells. One-sided hypergeometric test, no p -value adjustment for plotting. **d**, Enrichment plot showing the association of the YAP signature²² between P5 (m484) colon tumors and P1 (m4) colon tumors. Two-sided normalized enrichment score (NES) with Benjamini-Hochberg correction. Ranked list of genes (based on t statistics) are shown below in grey. **e - f**, Logo plots show the six most enriched transcription factor motifs and associated transcription factors in promoter (**e**) and enhancer (**f**) regions that open in P1 (m4) tumor epithelial cells. Adjusted p -values derived from one-sided hypergeometric tests with

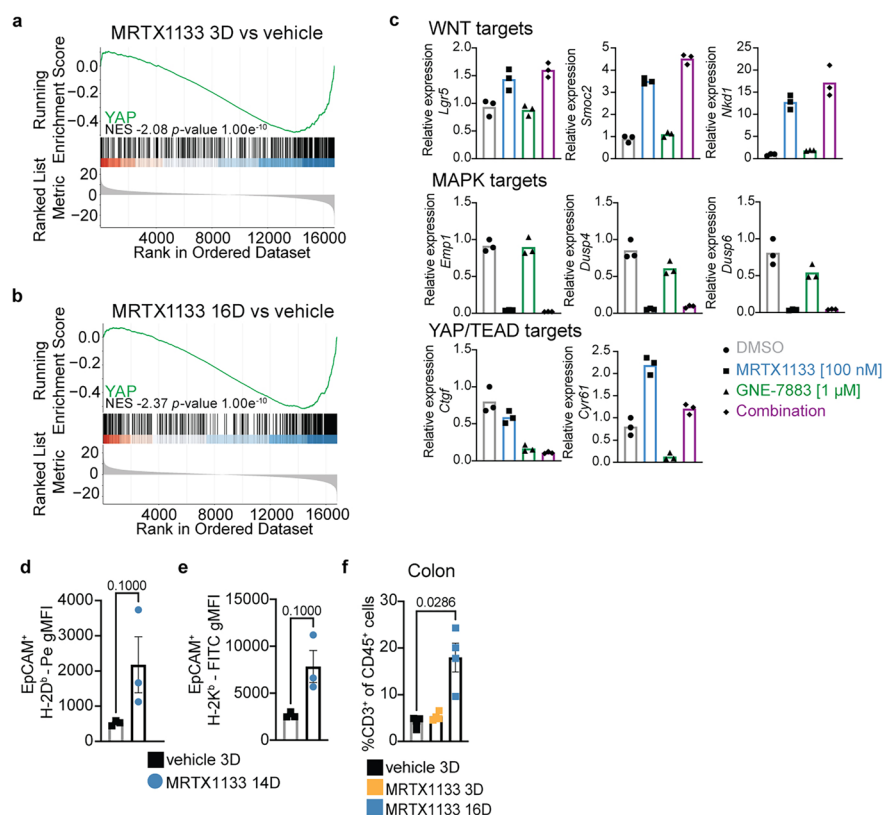
Benjamini-Hochberg false discovery rate (FDR). **g**, Top 60 transcription factor motifs enriched in regions with increased accessibility in P1 (m4) compared with P5 (m484) colon tumor epithelial cells. One-sided hypergeometric test, no p -value adjustment for plotting. **h**, Principal component analysis (PCA) plot of ATAC-seq profiles of P1 (m4) organoids, P5 (m484) organoids and P5 (m484) organoids treated for 8 days with one of the shown epigenetic inhibitors (EPZ004777 10 μ M; RG108 10 μ M; A-485 10 μ M; Panobinostat 50 nM). $n = 3$ replicates per treatment condition. **i**, Coverage plots showing library size-normalized ChIP-seq signal from ENCODE data at the human *EMP1* locus on chromosome 12. Tracks display BATF ChIP-seq in GM12878 cells (green, top; ENCODE ID - ENCSR000BGT), JUN-B ChIP-seq in A549 cells (brown, middle; ENCODE ID - ENCSR431LRW), and FOS ChIP-seq in MCF-7 cells (blue, bottom; ENCODE ID - ENCSR569XNP), with two individual replicates shown for each factor. The red box highlights the first exon of *EMP1*. Chr, chromosome.



Extended Data Fig. 5 | See next page for caption.

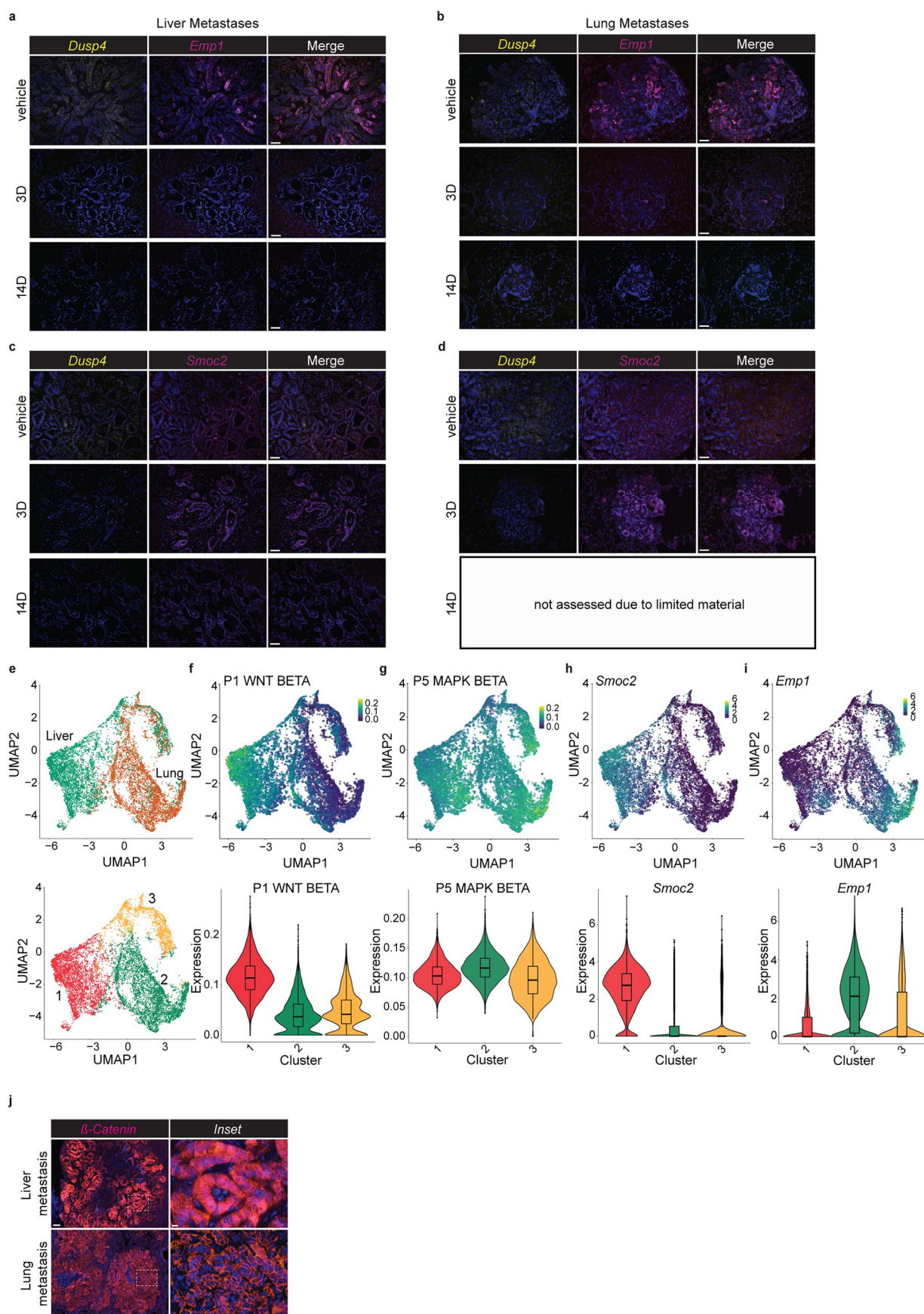
Extended Data Fig. 5 | Response of metastatic colon tumors to MRTX1133 treatment. **a**, Relative expression as determined by qRT-PCR analysis of MAPK signaling downstream targets *Dusp6* and *Etv4* in AKPS organoids harboring a *KRAS*^{G12D} mutation following 24 h treatment with increasing concentrations of the *KRAS*^{G12D} inhibitor MRTX1133. Two technical replicates per condition. Data are shown as relative expression normalized to *Rpl19*. Data displayed as mean. **b**, Western blotting analysis of MAPK signaling activity in P1 (m4) and P5 (m484) organoids after treatment with 100 nM MRTX1133 for 24 h. Data from a single experiment. **c**, Schematic of MRTX1133 treatment in mice. Mice received MRTX1133 (30 mg/kg i.p. twice a day) for either 3 or 16 days starting at 4.5 weeks post orthotopic injection of P5 (m484) into the colon. inj, injection. Illustration in **c** created in BioRender; Heinlein, M. <https://biorender.com/q31i735> (2025). **d**, Primary colon tumor weight at the indicated time points. Vehicle and MRTX1133 3 days $n = 4$ mice per group; MRTX1133 16 days $n = 5$ mice. **e**, EpCAM⁺ cells $\times 10^6$ per gram of colon tumor tissue as determined by FACS analysis on P5 (m484) colon tumors at indicated time points. $n = 4$ mice per group. **f**, PCA plot showing the distribution of RNA-sequencing samples on sorted colon tumor epithelial cells from P5 (m484) treated with vehicle (3 days; black) or MRTX1133 (3 days, yellow or 16 days, blue). $n = 4$ mice per group. **g - h**, Normalized gene set enrichment score of selected Wikipath gene signatures enriched in samples

with larger PC1 loadings (red) or smaller PC1 loadings (blue) (**g**) or PC2 loadings (red) or smaller PC2 loadings (blue) (**h**). Color gradients represent statistical significance. NES, normalized enrichment score. **i**, Enrichment plot showing the association of the P5 MAPK BETA signature (maroon) or P1 WNT BETA signature (teal) between colon tumors treated with MRTX1133 (3 days) relative to the vehicle treated controls. Two-sided normalized enrichment score (NES) with Benjamini-Hochberg correction. Ranked list of genes (based on t statistics) are shown below in grey. **j - k**, Standardized RNA expression and log₂ fold change of the P5 MAPK BETA (**j**) or P1 WNT BETA (**k**) target gene signature in tumor epithelial cells sorted from P5 (m484) colon tumors of mice treated with vehicle (3 days) or MRTX1133 (3 days or 16 days). Asterisks represent fold changes that are significant ($FDR \leq 0.05$). The exact FDR values for all fold changes are provided in Supplementary Table 1. $n = 4$ mice per group. veh, vehicle. **l - m**, Normalized gene set enrichment score of the CBC and RSC gene signatures¹⁸ in P5 (m484) tumor epithelial cells isolated from primary tumors treated with MRTX1133 or vehicle for 3 days (**l**) or 16 days (**m**). Numbers indicate statistical significance. For **d - f**: Each symbol represents a mouse. **d - e**, Data are presented as mean $\pm$ s.e.m. Groups were compared with Mann-Whitney U test (unpaired, no Gaussian distribution, two-tailed). P -values are shown in the corresponding panels.



Extended Data Fig. 6 | *KRAS*^{G12D} inhibition suppresses YAP signaling in metastatic colon tumors. **a - b**, Enrichment plot showing the association of the YAP signature²² between P5 (m484) colon tumors treated with MRTX1133 for 3 days (**a**) or 16 days (**b**) to the vehicle treated controls. Two-sided normalized enrichment score (NES) with Benjamini-Hochberg correction. Ranked list of genes (based on *t* statistics) are shown below in grey. **c**, Relative expression as determined by qRT-PCR analysis of WNT target genes *Lgr5*, *Smoc2*, *Nkd1* (top), MAPK target genes *Emp1*, *Dusp4*, *Dusp6*, (middle), and YAP/TEAD-associated genes *Ctgr* and *Cyr61* (bottom) in P5 (m484) organoids treated with DMSO, the *KRAS*^{G12D} inhibitor MRTX1133 (100 nM), the TEAD inhibitor GNE-7883 (1 μ M), or the combination (combo) for 48 h. Data are shown as relative expression

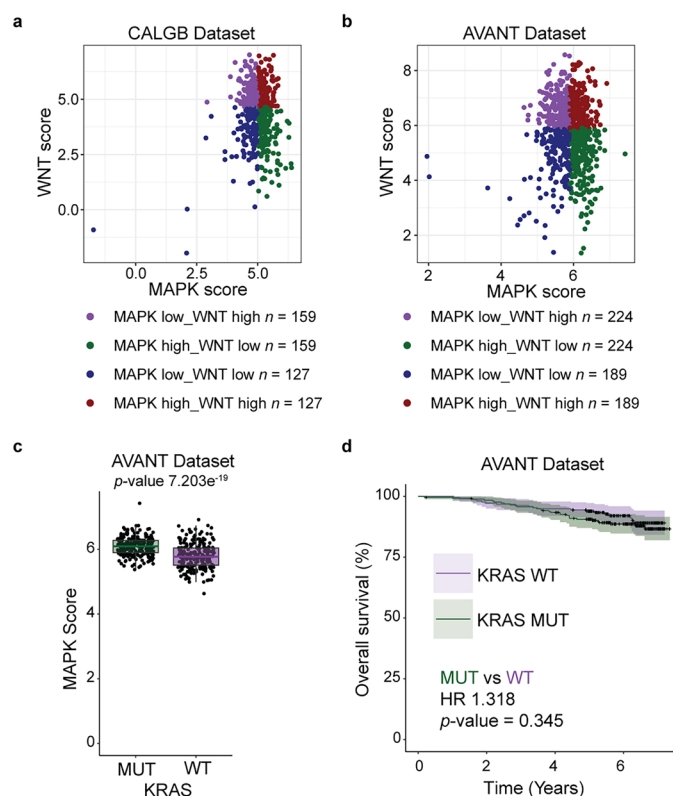
normalized to *Rpl19*. Symbols represent technical replicates (*n* = 3 per treatment condition) from a single experiment; bar graphs show mean. **d - e**, Geometric mean fluorescence intensity (gMFI) of H-2D^b (**d**) and H-2K^b (**e**) on EpCAM⁺ tumor epithelial cells from primary colon tumors after 14 days of MRTX1133 treatment or vehicle control. Mice were treated 3 weeks post orthotopic injection. *n* = 3 mice per group. **f**, Percentage of CD3⁺ T cells among CD45⁺ immune cells in primary colon tumors after 3 or 16 days of MRTX1133 treatment. Mice were treated 4.5 weeks post orthotopic injection. *n* = 4 mice per group. For **d - f**: Each symbol represents a mouse. Data are presented as mean $\pm$ s.e.m. Groups were compared with Mann-Whitney *U* test (unpaired, no Gaussian distribution, two-tailed). *P*-values are shown in the corresponding panels.



Extended Data Fig. 7 | See next page for caption.

Extended Data Fig. 7 | Response of metastatic lesions in the liver and lung to MRTX1133 treatment. **a - b**, Representative images of *Dusp4* (yellow) and *Emp1* (magenta) ISH and merged images (DAPI, blue) in liver (**a**) and lung (**b**) metastases from mice orthotopically injected with P5 (m484) organoids and treated 5 weeks after injection with vehicle (top), MRTX1133 for 3 days (middle), or 14 days (bottom). **c - d**, Representative images of *Dusp4* (yellow) and *Smoc2* (magenta) ISH and merged images (DAPI, blue) in liver (**c**) and lung (**d**) metastases under the same treatment conditions. Owing to limited material, lung metastases were not assessed after 14 days of treatment (**d**). Scale bars, 50 μ m. Sections from $n = 3$ mice per group (**a**, **c**, **d**) or $n = 2$ mice per group (**b**). **e**, UMAP plots of tumour epithelial cells isolated from liver and lung metastases of NSG mice orthotopically injected with AKVPSL organoids⁷, coloured by tissue of origin

(top) and cluster identity (bottom). **f - g**, UMAP plots showing P1 WNT BETA (**f**) and P5 MAPK BETA (**g**) signature scores (top), with corresponding violin plots showing score distributions across clusters (bottom). **h - i**, UMAP plots showing *Smoc2* (**h**) and *Emp1* (**i**) expression (top), with corresponding violin plots showing expression distributions across clusters (bottom). For **f - i** (bottom), box plots show the median (centre line), interquartile range (box), and 1.5 $\times$ interquartile range whiskers. **j**, Representative immunofluorescence images of β -catenin localisation in liver (top) and lung (bottom) metastases from NSG mice orthotopically injected with AKVPSL organoids⁷, with higher-magnification insets highlighting subcellular localisation. Sections from $n = 3$ mice per group. Scale bar, 75 μ m; inset scale bar, 10 μ m.



Extended Data Fig. 8 | Classification of CRC patients by MAPK and WNT pathway signatures. **a - b**, Scatter plots of MAPK canonical signature scores vs WNT canonical signature scores for individual patients in the CALGB (**a**) and AVANT (**b**) datasets. Patients are stratified relative to the median of each distribution: above-median MAPK and WNT (dark red), below-median MAPK and above-median WNT (purple), above-median MAPK and below-median WNT (green), and below-median MAPK and WNT (dark blue). **c**, MAPK target gene signature scores in KRAS-mutant (MUT; $n = 216$ patients) and KRAS wild-type (WT; $n = 258$ patients) colorectal cancer samples from the AVANT

dataset. Two-sided Wilcoxon-test. Box plots show individual data points, with the centre line indicating the median, the box representing the interquartile range, and whiskers extending to $1.5 \times$ the interquartile range. **d**, Kaplan–Meier overall survival curves for KRAS-mutant (green) and KRAS wild-type (purple) patients in the AVANT dataset. Hazard ratios and p -values were determined using Cox proportional hazards regression (two-sided) without multiple-testing correction. Solid lines represent Kaplan–Meier estimates, with shaded areas indicating 95% two-sided confidence intervals.

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Software and code

Policy information about [availability of computer code](#)

Data collection	Histological images acquisition: Bright-field whole-slide scanning was performed using Hamamatsu S360 microscope (NPD.view2). Immunofluorescent images were acquired on a Leica DM4 B with a Leica DFC 7000T camera (software Leica Application Suite X 3.4.2.18368). Flow cytometry and sorting: BD FACS Diva V9 and newer. Western Blot: Li-Cor Odyssey M Scanner Luminescence: Molecular Devices, Softmax Pro Version 7 qRT-PCR: Viiia7 real-time PCR system (Thermo Fisher Scientific) or CFX Bio-Rad Real-Time PCR Detection System (Bio-Rad). Sequencing machines: NovaSeq 6000 (Illumina); HiSeq4000 (Illumina),HiSeq 2500 (Illumina); NextSeq2000 (Illumina)
Data analysis	For data analysis the following software and platforms were used: Plotting and statistical analysis: Graphpad Prism Version 10.4.0 FACS analysis: FlowJo 10 Data analysis: The following tools and versions were used for analysis of whole exome sequencing data GATK (v4.2.6.1) LoFreq (v2.1.5.2) Strelka (v1.0.14) VEP (release 99) Control-FREEC (v11.6) The following tools and versions were used for analysis of bulk RNA-seq data

HTseqGenie (v4.35.0)
 GSNAP (2013-10-10-v2)
 edgeR (v4.0.15)
 Limma (v3.58.1)
 fgsea(v1.28.0)
 clusterProfiler (v4.10.0)
 Sva (v3.50.0)

The following tools were used for analysis of scRNA-seq data:

10X Cell Ranger
 UCell R package

The following tools and versions were used for analysis of bulk ATAC-seq data

Samtools (v1.9)
 Cutadapt (v2.5)
 ENCODE (v2.2.2)
 Bowtie2 (v2.3.4.3)
 Picard (v2.20.7) <https://broadinstitute.github.io/picard/>
 MACS2 (v2.2.4)
 IDR (v2.0.4.2)
 Diffbind (v3.12.0)
 HOMER (v4.11.1)
 ENCODE ATAC-seq pipeline (v2.2.2)

The following tools and versions were used for analysis of bulk RNA-seq and ATAC-seq integrated data

BETA (v1.0.7)

The following tools and versions were used for analysis of human patients survival data

survival (v3.5-8)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Data availability:

Mouse bulk RNA-sequencing, single cell RNA-sequencing and ATAC-sequencing data that support the findings of this study have been deposited in the Gene Expression Omnibus (GEO) under SuperSeries accession code GSE283937. Mouse Whole exome sequencing data has been deposited in SRA with accession number PRJNA1196349 and SRP559178.

Human RNA-sequencing data from the AVANT27 and CALGB28 studies are available in EGA under study ID EGAS50000001231. All other data supporting the findings of this study are available within the paper and its supplementary information.

Source data are provided with this paper.

The following figures have associated Source data: Fig 1a, b, d, e; Fig 2e, f, g, h; Fig 3f, g, h; Fig 5b, c, d, j, k, m, o; Ext data Fig 1c, e, f, g, h; Ext data Fig 2f, g, h, i, j, l, m, n, o; Ext Data Fig 3e; Ext Data Fig 5a, d, e; Ext Data Fig 6c, d, e, f.

Code availability:

No custom code was generated for this study. R code with intermediate data was deposited on GitHub https://github.com/Genentech/2025_heinlein_et al and is publicly available on Zenodo (<https://doi.org/10.5281/zenodo.17993514>).

Materials Availability

Requests for mouse organoid models and other reagents generated in this study will be available upon reasonable request. These materials will be distributed by Genentech upon completion of materials transfer agreement (MTA).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

AVANT - this clinical trial was previously published: [https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045\(12\)70509-0/fulltext](https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(12)70509-0/fulltext)
 826 patients with RNA-seq data were used in this analysis of which 335 were female and 491 male.

CALGB - this clinical trial study was previously published: <https://jamanetwork.com/journals/jama/fullarticle/2632502>
 A total of 1137 patients were included in the study of which 440 [39%] were women. 572 patients with RNA-seq data were used in this manuscript. However, details on sex/gender for the patients with RNA-seq data is not available.

Reporting on race, ethnicity, or other socially relevant groupings

AVANT:

Race/ethnicity of patients with available RNA-seq data included in this study was determined by self-report: American Indian or Alaskan Native: 4; Asian 22; Black 10; White 770; Other 20.

CALGB:

Race/ethnicity for the main study was determined by self-report. However, details on race/ethnicity for the patients with RNAseq data used in this manuscript is not available.

Population characteristics

AVANT:

Eligible patients were 18 years or older with histologically confirmed stage III or high-risk stage II colon carcinoma (defined by the American Joint Cancer Committee/Union Internationale Contre le Cancer). 3451 patients from 330 centers in 34 countries worldwide were randomly assigned. 826 patients with RNA-seq data were used in this analysis. 216 were KRAS WT, 258 were MUT, the rest had no associated data.

CALGB:

Patients (≥18 years) enrolled at community and academic centers throughout the National Clinical Trials Network in the United States and Canada (November 2005-March 2012) with previously untreated advanced or metastatic colorectal cancer whose tumors were KRAS wt chose to take either the mFOLFOX6 regimen or the FOLFIRI regimen as chemotherapy and were randomized to receive either cetuximab (n=578) or bevacizumab (n=559). Among 1137 patients (median age, 59 years; 440 [39%] women), 1074 (94%) of patients met eligibility criteria. Cure was the intent for 17.1% (99 of 578) of patients in the cetuximab group vs 13.2% (74 of 559) in the bevacizumab group. 572 patients with RNA-seq data were used in this analysis. Treatment arm A is Bevacizumab + Chemo 232 patients, B is Cetuximab + Chem 231 patients, and C is Bevacizumab + Cetuximab + Chemo 127 patients.

Recruitment

AVANT:

Eligible patients were 18 years or older with histologically confirmed stage III or high-risk stage II colon carcinoma (defined by the American Joint Cancer Committee/Union Internationale Contre le Cancer). Surgery with curative intent was done 4–8 weeks before randomisation. Key exclusion criteria included: evidence of remaining tumour; carcinoembryonic antigen levels of more than 1.5 times the upper normal limit after surgery; previous antiangiogenic treatment; major surgical procedure, open biopsy, or significant traumatic injury less than 28 days before study treatment; and abnormal haematological, liver, or renal function. From Dec 2004 to June 2007, 3,451 patients were randomized on the treatment arm (1151 received FOLFOX 4, 1155 received bevacizumab-FOLFOX4, and 1145 received bevacizumab-XELOX).

CALGB:

Patients (≥18 years) enrolled at community and academic centers throughout the National Clinical Trials Network in the United States and Canada (November 2005-March 2012) with previously untreated advanced or metastatic colorectal cancer whose tumors were KRAS wt chose to take either the mFOLFOX6 regimen or the FOLFIRI regimen as chemotherapy and were randomized to receive either cetuximab (n=578) or bevacizumab (n=559). The last date of follow-up was December 15, 2015. Patients were enrolled at centers across the National Cancer Trials Network in the United States and Canada. Eligible patients had pathology-documented untreated locally advanced or metastatic colorectal cancer, although measurable disease (tumor that could be quantified) was not required. Patients had to be candidates for either mFOLFOX6 or FOLFIRI regimens without known central nervous system metastases or grade II or greater peripheral neuropathy. In addition, hypertension had to be well controlled (blood pressure <160/90 mm Hg with treatment) and there could be no concurrent congestive heart failure. Therapeutic anticoagulation was permitted as long as the patient was therapeutic on a stable dose of anticoagulant.

Ethics oversight

AVANT - NCT00112918

This study was done in accordance with the declaration of Helsinki. Protocol approval was obtained from the ethics review committees or institutional review boards at participating sites. Patients provided written informed consent before study participation.

CALGB - NCT00265850

Institutional review board approval was required at all participating centers and all participating patients provided written informed consent.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Sample sizes in this study were determined based on a combination of prior experimental experience, published literature, and practical considerations. No formal statistical methods were used to predetermine sample sizes. Instead, sample sizes were chosen empirically according to previous knowledge of the variation in the experimental set up and to ensure sufficient power to detect biologically relevant

differences and to ensure robust and reproducible results across experiments.

Mouse studies: no statistical method was used to predetermine sample sizes; however, our sample sizes were similar to those reported in previous publications. For animal studies, this approach also adhered to the principles of the 3R system: Replacement, Reduction, and Refinement.

Covariates were not included in the statistical analyses. This was due to the predefined experimental design, limited sample sizes, and the use of controlled experimental conditions (including consistent sex, genetic background, treatment timing, and housing conditions), which minimized potential confounding. Accordingly, statistical analyses were performed without adjustment for covariates.

Data exclusions	Mice that did not grow tumors (i.e. due to a failed injection), that had to be euthanized before the defined endpoint or grew tumors outside the colon and into the peritoneal space (i.e. non-physiologically relevant) were excluded from downstream analysis. For in vitro experiments, no data were excluded.
Replication	All attempts at replication were successful. Multiple biological independent experiments were performed wherever feasible as described in the figure legends. In vivo experiments comparing P1 and P5 including drug treatments were repeated at least twice, unless otherwise stated. Sequencing experiments were performed once but with at least 3 biological repeats each, with the exception of whole exome sequencing for which only 1 replicate per organoid line was used. Tissue sections from three individual animals per organoid line or treatment group were stained for all H&E and in-situ hybridization staining to ensure reproducibility, with the exception for lung metastasis data in Extended Data 7 b and d.
Randomization	For in vivo drug studies, control and experimental animals were co-housed until group out. Animals were assigned to treatment groups in a partially randomized manner, so that treatment groups would consist of animals with similar size colon tumors, as determined by anal swelling, and no differences in mean body weight at the start of the experiment. For other experiments, randomization was not required.
Blinding	Data collection and analyses for in vivo experiments were not performed blinded to the conditions of the experiments for animal welfare reasons.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

FACS stainings:

Fc block (2.4G2; 1:500, BD, 553142)
 anti-mouse CD45 Brilliant Violet 605TM (30-F11, Biolegend 103140, 1:100),
 anti-mouse CD45 Alexa Fluor 700 (30-F11, Biolegend 103128, 1:100),
 anti-mouse CD326 APC (G8.8, Biolegend 118214, 1:100),
 anti-mouse CD31 PE/Cyanine 7 (390, Biolegend 102418, 1:100)
 anti-mouse CD3 PE/Cyanine 7 (17A2, Biolegend 100220 1:100),
 anti-mouse H-2Ld/H-2Db PE (28-14-8, Biolegend 114507; 1:200),
 anti-mouse H-2Kb FITC (AF6-88.5, Biolegend 116506; 1:200).

Western Blot analysis:

SMAD4 (D3R4N, 1:1000, Cell Signaling 46535),
 pMEK (41G9, 1:1000; Cell Signaling 9154),
 MEK (25/MEK1, 1:1000; BD 610122),
 pERK (D13.14.4E, 1:1000; Cell Signaling 4370),
 ERK (137F5, 1:1000; Cell Signaling 4695),
 pRSK (E238, 1:1000; Abcam ab32413),
 RSK (32D7, 1:1000; Cell Signaling 9355),
 DUSP6 (EPRI29Y, 1:1000, Abcam ab22081),
 beta-Actin (8H10D10, 1:1000; Cell Signaling 3700),
 beta-Tubulin (D3U1W, 1:1000, Cell Signaling 86298).

The secondary antibodies used:
 RDye 680LT Goat anti-Mouse IgG (Li-Cor, 926-68050)
 IRDye 800CW Goat anti-Rabbit IgG (Li-Cor, 926-32211).

Immunohistochemistry:
 mouse monoclonal anti- β -catenin antibody (clone 15B8, Abcam ab6301; 1:1000 dilution)

Validation

Pre-validated primary antibodies were purchased from Biolegend, Cell Signaling, Abcam or BD. Additionally, all primary antibodies used for FACS were tested and titrated in the relevant application to detect protein without cross-reactivity to non-specific proteins.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

Mouse organoid lines were generated from female mouse small intestinal tissue of either C57BL6/N wildtype mice or VilCreERT2 Apcfl/fl KrasLSL-Q61R/+ and VilCreERT2 Apcfl/fl KrasLSL-G12D/+ mice.

Authentication

For validation, organoids underwent sequencing to confirm CRISPR-Cas9 editing, western blot analysis or RNA-seq analysis.

Mycoplasma contamination

All organoid lines used in this study were routinely tested for mycoplasma contamination. All organoid lines used in this study tested negative for mycoplasma.

Commonly misidentified lines (See [ICLAC](#) register)

none

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

Wild-type female C57BL/6NJ (stock number 005304) and female NSG (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ) (stock number 005557) mice were purchased from the Jackson Laboratory. VilCreERT2 Apcfl/fl KrasLSL-Q61R/+ and VilCreERT2 Apcfl/fl KrasLSL-G12D/+ mice were bred at Genentech. Alleles used for this study were generated or licensed by Genentech from the appropriate institutions: VillinCreERT2 44, KrasLSL-G12D/+ 45, Apctm2Rak (designated Apcfl/fl)46, KrasLSL-Q16R 47. Apctm2Rak and KrasLSL-Q16R were maintained on a C57BL/6N background. KrasLSL-G12D/+ mice were backcrossed onto C57BL/6N and verified by SNP genotyping. Crosses with VillinCreERT2 resulted in a mixed C57BL/6N/J background. The mice were 7-16 weeks old at the start of the study.

Only female mice were used for in vivo studies since the organoids were generated from the small intestine of female mice.

Wild animals

No wild animals were used in this study.

Reporting on sex

Since the organoids used in this study were generated from female mice, and since this study used fully immunocompetent C57BL/6N mice, only female animals were used for the in vivo studies.

Field-collected samples

No field-collected samples were used in the study

Ethics oversight

Our research complies with all relevant ethical regulations. Animals were maintained in accordance with the Guide for the Care and Use of Laboratory Animals (National Research Council, 2011). Genentech is an Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC)-accredited facility. All animal activities in this research study were conducted under protocols 19-1847, 19-1976, 24-0259, 24-0694) approved by the Genentech Institutional Animal Care and Use Committee (IACUC) and the Laboratory Animal Resource Committee (LARC). These organizations provided guidance and approval for the study protocol.

Animals were euthanized at the humane endpoint, defined in accordance with institutional animal ethics committee guidelines. The maximal permitted tumor burden was defined as anal swelling with an ethical limit of 2.0 cm. The approved maximal tumor size and burden were not exceeded in any experiment. Additional humane endpoint criteria included >20 % body weight loss, severe lethargy, hunched posture, decreased grooming, severe diarrhea, rectal prolapse, or persistent bleeding.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Organoids were dissociated with accutase. Tumor tissues were minced separately into small pieces, placed in a 15-mL conical tube with 1-3 mL digestion medium (RPMI 1640 (Genentech, Inc) + 2 % heat-inactivated FBS (R&D Systems S12410H), 0.8 mg/mL Dispase (Life Tech. 17105041), 0.2 mg/mL collagenase P (Roche, 11249002001) or 0.1 mg/mL Liberase TL (Roche, 05401020001), and 0.1 mg/mL DNase I (Roche, 10104159001), and incubated at 37 °C for 15 minutes, with gentle resuspension every 5 minutes using a wide bore pipette. The supernatant was filtered (70 µm filter MACS Smart strainers 130-110-916) into RPMI + 10 % heat-inactivated FBS and placed on ice. Digestion was repeated until the whole tissue was brought into a single-cell suspension. Red blood cells were lysed using ACK (Genentech, Inc) buffer for 2–5 minutes at room temperature. Prior to cell surface staining with fluorescently labelled antibodies, cells were blocked with Fc block for 5 min at 4 °C. Cells were labelled with antibodies purchased from BioLegend. Live cells were identified by incubation with calcein blue (Invitrogen, C1429, 1:500) Sytox Blue (Thermo Fisher Scientific, S34857, 1:1000) or propidium iodide (Life Technologies, P3566) right before acquisition.
Instrument	Data were acquired on a Fortessa or Symphony (BD Biosciences). Cells were sorted on a Fusion or Aria (BD Biosciences)
Software	For collection: BD FACS diva (V9 and newer). Data were analyzed using FlowJo (Tree Star, 10.9.0)
Cell population abundance	Across experiments, sorted cells comprised of approximately 5-10% of all events detected. Post sort purity was above 99 % as determined by running sorted fractions on the same sorter that cells were originally sorted on.
Gating strategy	Gating strategy for bulk RNA-seq was as followed: FSC-A vs SSC-a: Exclude debris in the bottom left corner of plot SSC-H vs SSC-W: Only include single cells (doublet exclusion) FSC-H vs FSC-W: Only include single cells (doublet exclusion) Calcein Blue vs Propidium Iodide: gating on Calcein Blue positive PI negative cells to only obtain live cells CD45- BV605 vs Epcam- APC: Gating on Epcam positive CD45 negative to obtain tumor epithelial cells CD31-PeCy7 vs Epcam APC: Gating on Epcam positive CD31 negative to obtain tumor epithelial cells - this is the sorted population For scRNA-seq: FSC-A vs SSC-a: Exclude debris in the bottom left corner of plot SSC-H vs SSC-W: Only include single cells (doublet exclusion) FSC-H vs FSC-W: Only include single cells (doublet exclusion) Sytox Blue vs FSC-A: Gating on Sytox Blue negative cells to obtain live cells GFP vs RFP: gating on all RFP+ cells to obtain tumor epithelial cells

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.